Thermally induced deformation and conjugate heat transfer in a channel with two-way thermoelastic coupling

Mathematical formulation, reconciled analytical development, and verification plan

Pre-implementation research document — *International Journal of Numerical Methods for Heat & Fluid Flow* (target venue)

Version of 18 October 2026

Abstract

A mathematical formulation is developed for steady conjugate heat transfer in a two-dimensional channel whose upper wall contains a finite thermoelastic segment heated on its outer face. Heat conduction through the wall establishes a through-thickness temperature gradient; the associated thermal eigen-curvature deflects the segment into the flow passage; the deformed passage modifies the velocity and temperature fields; and the resulting interfacial heat flux closes the coupling back onto the wall temperature. The model couples incompressible laminar flow, advection–diffusion heat transfer, steady conduction, and linear isotropic thermoelasticity under plane strain, with traction and heat-flux continuity imposed on a common deformed interface, the wall equation being carried to the pinned beam–column form at finite membrane force. The plane-strain constitutive reduction, mean-stress identity, free thermal curvature, support-dependent bowing and thermal-buckling limits, and the pressure-datum-consistent relative magnitude of hydraulic and thermally driven deflection are derived explicitly for the present configuration. A baseline operating point is evaluated from first principles, the governing dimensionless groups are identified from the equations on a single characteristic-length basis, and a conforming-mesh finite-element formulation with streamline-stabilized transport and a deformed-domain fixed-point iteration is proposed together with a verification hierarchy based on analytical solutions and published benchmark data. The document distinguishes established theory, analytical results derived for this configuration, and quantities that can only be obtained from the future computations; no simulation results are reported. Discrepancies identified and reconciled during preparation of the formulation — each with its resolution basis, and the few items that remain explicitly unresolved — together with the conditions gating coupled production computations, are collected in the appendices and in the accompanying audit archive.

Contents

1 Introduction and physical motivation **3**
- 1.1 Problem statement . . . 3
- 1.2 Physical motivation . . . 3
- 1.3 Scope of the present study . . . 4
- 1.4 Relation to existing work . . . 4
- 1.5 Organisation of the document . . . 4

2 Mathematical model and modelling assumptions **5**
- 2.1 Geometry and reference configuration . . . 5
- 2.2 Modelling assumptions . . . 5
- 2.3 Sign and traction conventions . . . 7
- 2.4 Material data . . . 7
- 2.5 Notation . . . 8

3 Governing equations and constitutive relations **9**
- 3.1 Mass conservation in the fluid . . . 9
- 3.2 Momentum conservation in the fluid . . . 9
- 3.3 Energy conservation in the fluid . . . 10
- 3.4 Heat conduction in the solid . . . 10
- 3.5 Linear thermoelastic constitutive relations and kinematics . . . 10
  - 3.5.1 Plane-strain closure and out-of-plane stress . . . 10
- 3.6 Elastic equilibrium . . . 11
- 3.7 Mean stress and the spatial mean-stress identity . . . 11
- 3.8 Reduced wall equation for design estimates . . . 11
- 3.9 The deformed fluid domain . . . 11
- 3.10 Constitutive completeness . . . 11

4 Derivation of the plane-strain thermoelastic formulation **13**
- 4.1 Plane-strain reduction of the constitutive law . . . 13
- 4.2 Mean-stress identity . . . 13
- 4.3 Reference equilibrium states (sanity checks) . . . 14
- 4.4 Free thermal curvature of the strip . . . 14
- 4.5 Pinned–pinned thermal bowing . . . 14
- 4.6 Thermal membrane force and pinned–pinned buckling . . . 15
- 4.7 Beam–column bowing at finite membrane force . . . 15
- 4.8 Hydraulic loading and the deformation ratio . . . 16
- 4.9 Time scales and the steady-state reduction . . . 17

5 Boundary and interface conditions **18**
- 5.1 Interface conditions on the fluid–solid boundary . . . 18
- 5.2 Exterior boundary conditions . . . 18
- 5.3 Interface data flow in the coupled problem . . . 19

6 Nondimensionalization and characteristic parameters **20**
6.1 Groups from the fluid equations . . . 20
6.2 Groups from the solid and interface equations . . . 21
6.3 Admissibility bounds on the regime . . . 21
6.4 Dependence on yet-unfrozen data . . . 21

7 Baseline operating conditions and analytical calculations **22**
7.1 Baseline calculation table . . . 22
7.2 Interpretation . . . 24
7.3 Buoyancy disposition at the baseline . . . 25

8 Proposed numerical formulation and solution procedure **26**
8.1 Spatial discretization . . . 26
8.2 Weak formulation and monolithic interface cancellation . . . 27
8.3 Solution procedure for the steady problem . . . 27
8.4 Mesh convergence and quality controls . . . 28
8.5 Post-processing definitions . . . 28
8.6 Computational feasibility note . . . 28

9 Verification and validation methodology **29**
9.1 Benchmark hierarchy . . . 29
9.2 Benchmarks explicitly not used . . . 31
9.3 Cross-component verification items . . . 31
9.4 Staged implementation decisions . . . 31
9.5 Independent replication . . . 32

10 Literature review, research gap, and proposed contribution **33**
10.1 Lines of research neighbouring the present model . . . 33
10.2 Research gap and its stated bounds . . . 33
10.3 Intended contribution of the proposed work . . . 34

11 Limitations and concluding remarks on the formulation **35**
11.1 Model limitations . . . 35
11.2 Concluding remarks . . . 36

A Notational conventions and arithmetic calculations **37**
A.1 Notation and units . . . 37
A.2 Arithmetic cells (independent recomputation) . . . 38
A.3 Dimensionless dictionary . . . 38

B Audit record: discrepancies, evidence status, and open checks **40**
B.1 Discrepancy register . . . 40
B.2 Consolidated evidence status . . . 43
B.3 Open checks (with named action items) . . . 44
B.4 Bibliographic integrity statement . . . 44
B.5 Staged decisions carried from the audit . . . 44
B.6 Second reconciliation pass: scope and disposition . . . 44
B.7 Stage-3 synchronization: beam–column model classification . . . 45

Chapter 1

Introduction and physical motivation

1.1 Problem statement

Consider steady laminar forced convection in a two-dimensional slit channel. A finite segment of the upper wall is not rigid: it consists of a thin thermoelastic layer whose outer face is heated by a uniformly applied heat flux, as realised, for example, by a film heater. Conduction across the layer produces a through-thickness temperature difference whose magnitude is governed by the wall conductivity and thickness. Linear thermoelasticity associates with this gradient a free thermal curvature; with the segment ends pinned, the segment bows toward the interior of the passage. The resulting displacement redistributes the flow cross-section, alters the near-wall velocity and temperature gradients, and therefore changes the interfacial convective heat flux. That flux, in turn, is one of the boundary data of the conduction problem in the wall. The wall temperature, the deformation, and the fluid thermal field are thus coupled in both directions, and a consistent solution requires their simultaneous determination.

The present document formulates this coupled problem mathematically, derives the reduction implied by plane strain and by the thin-wall geometry, evaluates a baseline operating point analytically, and specifies a numerical formulation and a verification hierarchy for its eventual computation. No simulation results are reported; the distinction between established theory, results derived here for this configuration, and quantities that remain to be computed is maintained throughout.

1.2 Physical motivation

Wall-guided micro heat exchangers and microreactors are increasingly fabricated from polymers and polymer composites, which combine low thermal conductivity with a comparatively high coefficient of thermal expansion. For the same imposed heat flux, such walls sustain through-thickness temperature drops orders of magnitude larger than metallic walls of equal thickness, and their free thermal curvature is correspondingly larger (the comparison is quantified in Chapter 7). When the wall thickness is comparable to the channel height, the associated deformation is not a perturbation of the passage geometry but a modification of it, and heat-transfer predictions that assume a fixed geometry become geometry-dependent in a thermally self-consistent way.

Deformation of flow passages by thermal loading is well known from thermoelastic-contact and regenerative-cooling analyses, in which the deformation is typically computed a posteriori from a frozen flow solution, i.e. in one direction of coupling. In the configuration considered here, the deformation is itself the principal quantity of interest and feeds back on the convective boundary condition that produces it. A number of natural modelling questions follow and motivate the present work: under which combinations of wall material, geometry, and hydraulic operation is the thermal bowing comparable to the channel height; what bounds does thermal buckling of the pinned segment impose on admissible heating; how does the interaction behave under imposed

flow rate relative to imposed pressure drop; and how large are the residual stresses carried by the wall.

1.3 Scope of the present study

The analysis is confined to a two-dimensional geometry, steady state, laminar incompressible flow, and linear isotropic thermoelasticity under plane strain, with the wall equation extended to the pinned beam–column form at finite membrane force. The baseline treats imposed heat flux on the outer face of the deformable segment; a variant with imposed outer-face temperature is retained as a separately documented operating regime. The solid is modelled by two-dimensional elasticity and steady conduction; the reduced wall equations are used for design estimates and benchmark comparisons. Buoyancy is disposed of at the baseline by a length-consistent Richardson-number estimate (three orders of magnitude below the mixed-convection range), with a regime guard that re-opens the question should a parameter study drive the number upward, as discussed in Chapters 7 and 11.

Transient coupling, mesh dynamics, and the associated discrete geometric conservation-law requirements [13] are outside the present scope and are recorded as the conditions of a possible start-up study. Likewise, creep of the polymer wall, manufacturing tolerances, and contact resistance at the interface are excluded from the baseline and are listed among the model limitations.

1.4 Relation to existing work

Fluid–structure interaction with heat transfer has been studied for cavity and channel geometries in which the deformation is driven by fluid traction [4, 5, 14], and fully nonlinear thermo-fluid–structure interaction of flexible cantilevers in a differentially heated cavity has been computed [7]. In the configuration studied here, the deformation mechanism is instead the wall's own thermal eigen-curvature, the flow is a confined forced-convection slit at moderate Reynolds number, and the coupling closes through the conjugate thermal boundary condition. A detailed comparison, the resulting positioning of the intended contribution, and the limits of that positioning imposed by the available literature search are developed in Chapter 10.

1.5 Organisation of the document

Chapter 2 fixes the geometry, assumptions, sign conventions, and material data. Chapter 3 states the governing equations and the constitutive relations together with the principles from which they follow. Chapter 4 derives the plane-strain reduction and the analytical relations used for design estimates. Chapter 5 assembles the boundary and interface conditions. Chapter 6 identifies the dimensionless groups and the admissibility bounds of the regime. Chapter 7 evaluates the baseline operating point in full arithmetic detail. Chapter 8 proposes the finite-element formulation and solution procedure. Chapter 9 specifies the verification and validation methodology. Chapter 10 reviews the related literature and states the proposed contribution and its bounds. Chapter 11 collects the limitations. Appendices contain the notation, the arithmetic calculations in independently checkable cells, a dimensionless dictionary, and the audit record (discrepancy register, consolidated evidence status, unresolved source checks, and the conditions attached to the staged implementation decisions).

Chapter 2

Mathematical model and modelling assumptions

2.1 Geometry and reference configuration

The configuration is two-dimensional. Reference coordinates $X = (X, Y)$ label material points of the solid and the undeformed fluid domain. The fluid reference domain, $\Omega_f^0 = \{(X, Y) : -L_e \leq X \leq L + L_o,\ 0 \leq Y \leq H_0\}$, consists of a rigid entrance section $\Omega_e = (X \in [-L_e, 0])$, a section $\Omega_d = (X \in [0, L])$ facing the deformable wall, and a rigid exit section $\Omega_o = (X \in [L, L + L_o])$, all of undeformed height H_0. The only deformable body is the solid strip $\Omega_s^0 = [0, L] \times [H_0, H_0 + d]$ of thickness d spanning the full out-of-plane depth. Its lower face $Y = H_0$ shares the fluid boundary; its upper face $Y = H_0 + d$ receives the imposed thermal load. The strip is embedded in a rigid casing and is pinned at both ends, $X = 0$ and $X = L$: the displacement vanishes there while the bending moment vanishes (Assumption A9). Figure 2.1 shows the configuration, Table 2.1 records the design point.

The deformed interface is defined by the solid displacement field $\mathbf{d}$ on the reference interface,

$$\Gamma_{\text{if}} = \{x = X + \mathbf{d}(X) :\ X \in [0, L] \times \{H_0\}\}, \tag{2.1}$$

and the steady fluid domain Ω_f is obtained by extending the interface displacement into Ω_f^0 through a mesh-extension map $\widehat{\mathbf{w}}$ specified in Section 3.9. The fluid equations are posed on Ω_f; the solid equations on the reference configuration, consistent with small strains (Assumption A4).

2.2 Modelling assumptions

Each assumption is stated with its justification and its range of validity; the quantitative checks supporting the numerical bounds are carried out in Chapter 7.

A1 *Two-dimensional geometry and plane strain.* The out-of-plane dimension is large compared with H_0 and d, and loading is uniform along it, so the flow and temperature fields are two-dimensional and the solid is in plane strain, $\varepsilon_{zz} = 0$. Plane strain, rather than plane stress, is the closure appropriate to a wide bonded wall [8, 17]; its consequences for the constitutive reduction are derived in Chapter 4.

A2 *Newtonian, constant-property fluid.* The working fluid is water at near-ambient temperature, with constant density ρ, viscosity μ, specific heat c_p, and conductivity k_f. The self-consistency of constant properties is established a posteriori: the computed fluid-side temperature excursions at baseline are of order 1–3 K (Section 7.1), over which the property variation remains below a few per cent. The influence of buoyancy is examined separately through the Richardson number (Sections 7.3, 11.1).

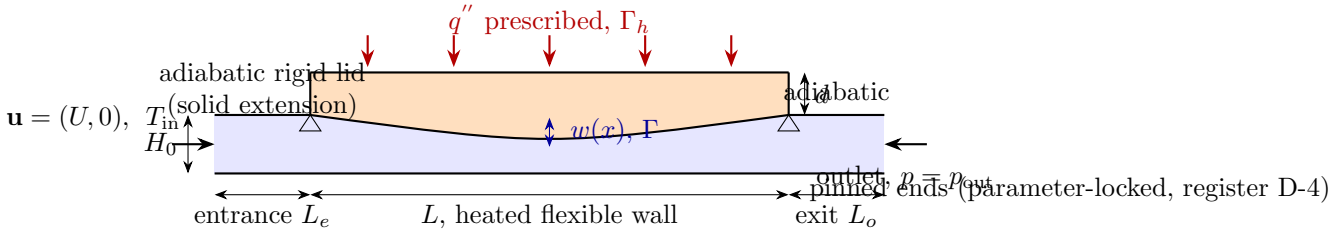


Figure 2.1. Configuration of the coupled problem (schematic, not to scale). Rigid entrance and exit sections of the channel; a thermoelastic segment of length L and thickness d, pinned at its ends and heated on its outer face Γ_h^0; the conjugate interface Γ_{if} between fluid and solid; a rigid, adiabatic lower wall. The interface deflection w is measured positive into the channel (Figure 2.2).

Table 2.1. Geometry and operating point used as the baseline design. Values marked *design* are free choices of the study; the derived hydraulic diameter uses the slit value $D_h = 2H_0$.

Quantity	Symbol	Value	Basis
Channel height	H_0	5.0×10^{-4} m	design
Hydraulic diameter	$D_h = 2H_0$	1.0×10^{-3} m	derived
Heated, deformable length	L	3.0×10^{-2} m	design
Entrance / exit lengths	L_e, L_o	1.0×10^{-2} m; 2.0×10^{-2} m	design
Wall thickness	d	3.0×10^{-3} m	design
Mean inlet velocity	U	5.0×10^{-2} m s^{-1}	design
Imposed heat flux (baseline regime)	q''	5.0×10^{3} W m^{-2}	design
Inlet and reference temperatures	$T_{\text{in}}, T_{\text{ref}}$	to be fixed at data freeze	temperature differences only enter

A3 *Homogeneous isotropic linear thermoelastic solid.* The wall is a PEEK-class polymer with constant E, ν, α, k_s, ρ_s, c_s. Deviations (grade-to-grade variability, mild anisotropy of extruded stock) are treated by the sensitivity requirements of Chapter 9; the numerical values used below are datasheet-representative and are to be frozen against a specific datasheet before implementation (Appendix B, item C-6).

A4 *Small strains, moderate rotations.* The strain measures must remain small: the bending fibre strain, the arc-stretching membrane strain, and the total mechanical surface strain. Linear kinematics alone is *not* sufficient for the pinned, axially constrained strip, because the thermal membrane force interacts with the transverse deflection at leading order; the adopted kinematics is therefore the reduced (frozen-force) von Kármán beam–column level of Section 4.7, which reduces to the linear relation when the membrane force is negligible. At the baseline the bending fibre strain is 9.9×10^{-4}, the maximum total mechanical surface strain is $\approx 5.4 \times 10^{-3}$, the arc-stretching term is $\approx 1.8 \times 10^{-4}$, and the maximum slope is 0.03 rad (Section 7.1); all are far below the order-one failure limits of the hypothesis. Geometric admissibility is expressed through these strain and slope measures together with a mesh bound $|w|/H_0 \leq 0.5$ with positive element Jacobians, rather than through a purely geometric deflection ratio alone (Appendix B, register D-11, D-15).

A5 *Steady flow and energy; quasi-static solid mechanics.* The structural wave time across the wall is seven orders of magnitude shorter than the slowest transport time (Section 4.9), so the mechanical problem is quasi-static and inertia is omitted from (3.7).

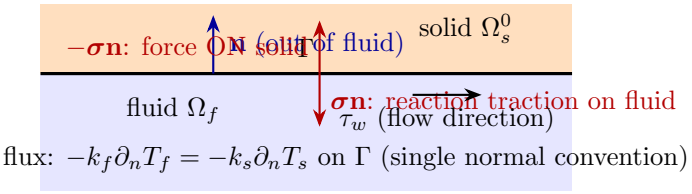


Figure 2.2. Interface conventions. A single unit normal $\mathbf{n}$ (fluid $\rightarrow$ solid) is used in the traction equality (5.2), in the flux equality (5.4), and in all variational boundary terms of Section 8.2.

A6 *Laminar flow.* The design keeps $\mathrm{Re}_{D_h} \leq 100$ (baseline value 50), well below the transition range for plane channels, which justifies (3.2).

A7 *Negligible viscous dissipation.* The Brinkman number at baseline is of order 10^{-6} (Chapter 7); the dissipation term is retained as a re-inclusion option with the rule stated at (3.3).

A8 *Perfect thermal contact at the interface.* No contact resistance is included at Γ_{if}; temperature and heat flux are continuous ((5.3)–(5.4)).

A9 *Ideal pinned supports and rigid casing.* The strip ends have zero displacement, zero applied moment, and zero heat flux; the surrounding casing is taken rigid and adiabatic, and its interior pressure equals the outlet datum of condition B2, which fixes the load reference of Section 4.8. Implementation note: the pins are imposed on the full end edge ($X = 0, L$), which differs from ideal point pins by Saint-Venant end layers of width $O(d)$; at the baseline aspect ratio $L/d = 10$ this contributes a few per cent of global stiffening, and a one-time numerical check (full-edge versus corner pins) is part of the verification programme (Chapter 9).

A10 *Uniform imposed flux as the baseline thermal load (regime N).* A uniform q'' models a film heater and fixes the total heat input per unit depth, $Q' = q''L$, which is the quantity entering the global enthalpy balance (Section 8.4). A second regime (regime D, imposed outer-face temperature T_h) is retained as a separately labelled study; comparisons across regimes are meaningful only at matched total heat input.

2.3 Sign and traction conventions

A single interface normal $\mathbf{n}$ is used throughout, directed from the fluid into the solid. Pressure on the wall is positive in compression; Cauchy stress in the solid is positive in tension. The interface deflection is denoted w and is measured *from the reference interface into the channel* (hence $w = -d_2$ in terms of the upward-positive solid displacement d_2). Wall curvature is taken positive when the wall bows into the channel; within the small-slope assumption this gives the design relation $\kappa = -w''$, and a constant eigen-curvature $\kappa_{\mathrm{th}} > 0$ produces $w = \kappa_{\mathrm{th}} x(L - x)/2 \geq 0$ (Section 4.5). Fluid heat flux along $+\mathbf{n}$ carries heat out of the fluid. Using one normal on both sides of the traction equality is what renders the interface boundary terms in the weak statement of Section 8.2 exactly antisymmetric, so that they cancel in a monolithic assembly without modelling error.

2.4 Material data

Table 2.2 lists the property values used in all calculations. Fluid values are standard near-ambient water data. Solid values are representative of unfilled PEEK; they are marked accordingly, because they must be frozen against a chosen datasheet before the numerical implementation

(Appendix B, item C-6), and the derived baseline quantities that depend on them are reported with that dependency stated.

Table 2.2. Material properties used in the calculations. "freeze pending" = datasheet value to be fixed before implementation (Appendix B, item C-6).

Property	Symbol	Value	Units	Remark
Water density	ρ	1.0×10^3	$\mathrm{kg\,m^{-3}}$	standard value
Water dynamic viscosity	μ	1.0×10^{-3}	$\mathrm{Pa\,s}$	standard value
Water specific heat	c_p	4.18×10^3	$\mathrm{J\,kg^{-1}\,K^{-1}}$	standard value
Water conductivity	k_f	0.60	$\mathrm{W\,m^{-1}\,K^{-1}}$	standard value
Water expansion coefficient	β_f	2.1×10^{-4}	$\mathrm{K^{-1}}$	freeze pending
Water kinematic viscosity	$\nu_{\mathrm{kin}} = \mu/\rho$	1.0×10^{-6}	$\mathrm{m^2\,s^{-1}}$	derived
PEEK Young's modulus	E	3.6×10^9	Pa	freeze pending
PEEK Poisson ratio	ν	0.40	–	freeze pending
PEEK expansion coefficient	α	5.0×10^{-5}	$\mathrm{K^{-1}}$	freeze pending
PEEK conductivity	k_s	0.25	$\mathrm{W\,m^{-1}\,K^{-1}}$	freeze pending
PEEK density	ρ_s	1.3×10^3	$\mathrm{kg\,m^{-3}}$	freeze pending
PEEK specific heat	c_s	1.1×10^3	$\mathrm{J\,kg^{-1}\,K^{-1}}$	freeze pending

2.5 Notation

Current and reference coordinates are $x = (x, y)$ and $X = (X, Y)$. Fluid unknowns are the velocity $\mathbf{u} = (u, v)$, pressure p, and temperature T_f; solid unknowns are displacement $\mathbf{d} = (d_1, d_2)$, temperature T_s, stress $\boldsymbol{\sigma}$, and strain $\boldsymbol{\varepsilon}$; $\theta = T_s - T_{\mathrm{ref}}$ denotes the temperature rise in the solid. The Lamé parameters are written λ and μ_L, where $\mu_L = G$ is the shear modulus (the subscript distinguishes it from the fluid viscosity). The bulk modulus is $K = \lambda + \frac{2}{3}\mu_L = E/[3(1-2\nu)]$, and the thermoelastic pressure coefficient is $\beta = (3\lambda + 2\mu_L)\alpha = 3K\alpha = E\alpha/(1-2\nu)$ [8]. A complete symbol table with units is given in Appendix A.1.

Chapter 3

Governing equations and constitutive relations

This chapter states the field equations of the coupled model, in each case with the conservation principle or constitutive theory from which the equation follows, the assumptions of Chapter 2 that make the stated form applicable, and the literature source of the general relation. Symbol definitions and units are collected in Chapter 2 and Appendix A.1; the boundary and interface conditions are assembled separately in Chapter 5.

3.1 Mass conservation in the fluid

For an incompressible continuum, conservation of mass reduces to the divergence-free condition on the velocity field [3],

$$\nabla \cdot \mathbf{u} = 0 \qquad \text{in } \Omega_f. \tag{3.1}$$

Under Assumption A2 the density is constant and (3.1) holds identically. The domain Ω_f is the *deformed* steady domain defined in Section 3.9; the deformation enters the fluid problem only through the domain geometry.

3.2 Momentum conservation in the fluid

Cauchy's momentum equation with constant density, constant viscosity, and no volume forces reduces to the steady Navier–Stokes system [3],

$$\rho\,(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u} \qquad \text{in } \Omega_f. \tag{3.2}$$

Assumptions A2 and A6 (constant Newtonian properties, laminar regime) make (3.2) well posed together with the boundary conditions of Chapter 5. Every term carries units $\mathrm{Pa\,m^{-1}}$. Buoyancy would enter through the Boussinesq body force $\rho \beta_f \theta_f \mathbf{g}$, where $\theta_f = T_f - T_{\mathrm{ref}}$ is the fluid temperature rise and β_f the volumetric expansion coefficient. The Richardson number computed in Chapter 7 on a length-consistent basis is 1.7×10^{-3}, so the Boussinesq buoyancy term is excluded by the standard order-of-magnitude argument: the buoyant forcing is three orders of magnitude below the inertial forcing. A Boussinesq body-force option is nevertheless kept dormant in the formulation (one weak-form term) so that any future operating point with a significantly larger fluid-side temperature excursion can be re-examined; for context on the opposite limit, in which stable stratification measurably modifies channel heat transfer at bulk Richardson numbers of order 10 and above, see [12]. The disposition is re-examined in the limitations chapter.

3.3 Energy conservation in the fluid

For a constant-property fluid, the thermal energy equation reduces to the steady advection–diffusion form [3]

$$\rho c_p\,(\mathbf{u}\cdot\nabla)T_f = k_f\nabla^2 T_f \qquad \text{in } \Omega_f. \tag{3.3}$$

Viscous dissipation $\mu\Phi$, with $\Phi = 2\mathbf{D}:\mathbf{D}$ and $\mathbf{D} = \frac{1}{2}(\nabla\mathbf{u}+\nabla\mathbf{u}^{\mathsf{T}})$ the rate-of-deformation tensor, is neglected under Assumption A7: at the baseline the Brinkman number is 1.0×10^{-6} (Section 7.1), and the dissipation term is required to be restored whenever a computed case reaches $\mathrm{Br} \geq 10^{-2}$. Each term has units $\mathrm{W\,m^{-3}}$.

3.4 Heat conduction in the solid

Steady conduction in the wall obeys [11]

$$\nabla\cdot(k_s\nabla T_s) = 0 \qquad \text{in } \Omega_s^0. \tag{3.4}$$

In the general theory of thermoelasticity the energy equation contains the deformation-heating source $-\beta T_0\,\mathrm{tr}\,\dot{\boldsymbol{\varepsilon}}$, where T_0 is the absolute reference temperature [8]. This source vanishes identically in any steady state ($\dot{\boldsymbol{\varepsilon}} \equiv 0$), so (3.4) is exact for the baseline. In a transient start-up study its omission would be an approximation controlled by the thermoelastic coupling number $\xi = \beta^2 T_0/[(\lambda+2\mu_L)\rho_s c_s]$, evaluated at the baseline in Section 7.1 ($\xi \approx 2\times10^{-2}$).

The solid domain Ω_s^0 is the reference configuration: under small strains (A4) the distinction between current and reference coordinates is negligible for the conduction problem. The wall thermal field feeds the mechanical problem through the constitutive relation (3.5), and is closed by the conjugate interface conditions of Chapter 5.

3.5 Linear thermoelastic constitutive relations and kinematics

The solid follows the Duhamel–Neumann extension of Hooke's law for an isotropic body with linear thermal expansion [8, 17],

$$\boldsymbol{\varepsilon} = \tfrac{1}{2}\big(\nabla\mathbf{d}+\nabla\mathbf{d}^{\mathsf{T}}\big), \qquad \boldsymbol{\sigma} = 2\mu_L\boldsymbol{\varepsilon} + \lambda(\mathrm{tr}\,\boldsymbol{\varepsilon})\mathbf{I} - \beta\theta\mathbf{I} \qquad \text{in } \Omega_s^0. \tag{3.5}$$

The assumptions are small strain (A4) and isotropy (A3). All stress components reported from the model are *total* Cauchy stresses, i.e. (3.5) is never split for reporting into a "mechanical" and a "thermal" part; the trace of (3.5) is examined in Section 3.7 and turns this convention into a checkable identity.

3.5.1 Plane-strain closure and out-of-plane stress

Under plane strain $\varepsilon_{zz} = 0$ (A1). Writing Hooke's law in compliance form with the free thermal strain,

$$\varepsilon_{zz} = \frac{1}{E}\big[\sigma_{zz} - \nu(\sigma_{xx}+\sigma_{yy})\big] + \alpha\theta = 0 \implies \sigma_{zz} = \nu(\sigma_{xx}+\sigma_{yy}) - E\alpha\theta, \tag{3.6}$$

which specialises the general plane-strain result [8, 17] to the present notation. The equivalence of (3.6) with the stiffness-form value $\sigma_{zz} = \lambda\,\mathrm{tr}\,\boldsymbol{\varepsilon} - \beta\theta$ is verified algebraically in Section 4.1. The out-of-plane stress must be retained in any field quantity that involves the stress trace, as the next section makes precise.

3.6 Elastic equilibrium

Neglecting inertia (A5) and volume forces, linear-momentum balance in the solid reduces to

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \qquad \text{in } \Omega_s^0. \tag{3.7}$$

Together with (3.5) and the stress boundary conditions of Chapter 5, (3.7) constitutes the classical displacement formulation of isothermal thermoelasticity [8]; elimination of stress gives the Navier equation with the thermal body-force-equivalent term $-\beta\nabla\theta$.

3.7 Mean stress and the spatial mean-stress identity

The mean (hydrostatic) Cauchy stress $\sigma_m = \operatorname{tr}\boldsymbol{\sigma}/3$ is used to characterise the compressive membrane state of the wall. Two equivalent expressions follow from (3.5) and (3.6) and are derived in Section 4.2:

$$\sigma_m = \frac{1+\nu}{3}\big(\sigma_{xx}+\sigma_{yy}\big) - \frac{E\alpha}{3}\,\theta = K\big(\operatorname{tr}\boldsymbol{\varepsilon} - 3\alpha\theta\big), \tag{3.8}$$

where the second form identifies mean stress with bulk modulus $\times$ mechanical volumetric strain. The two forms must agree identically at every point; this identity is used as an algebraic consistency check in the post-processing of the future solver, and only complete thermoelastic stress components may enter the first form. An immediate consequence for sanity checks (Section 4.3) is that under plane strain even a *free* uniformly heated strip carries $\sigma_m = -E\alpha\theta/3 \neq 0$, because $\varepsilon_{zz} = 0$ is itself a constraint that generates $\sigma_{zz} = -E\alpha\theta$; this corrects the three-dimension-free expectation $\sigma_m = 0$ recorded in an earlier draft of the formulation (Appendix B, register D-2).

3.8 Reduced wall equation for design estimates

For design estimates and beam-level benchmarks, the strip is additionally approximated as a thin Euler–Bernoulli plate strip of flexural rigidity per unit depth $D_s = Ed^3/[12(1-\nu^2)]$ [17], governed by

$$D_s\, w''''(x) = -\,p(x), \qquad w(0) = w(L) = w''(0) = w''(L) = 0, \tag{3.9}$$

with w measured positive into the channel (Section 2.3) and $p(x)$ the fluid pressure acting on the wall, pushing the wall *away* from the channel; the sign convention is fixed so that the pinned–pinned thermal bowing of Section 4.5 is positive. Equation (3.9) is used *only* for design estimates and benchmark construction; the production model is the two-dimensional system (3.4)–(3.7), of which (3.9) is the standard leading-order reduction for $d/L \ll 1$ [17].

3.9 The deformed fluid domain

The fluid problem is posed on the domain Ω_f occupied by the fluid in the converged steady configuration. Given the solid solution on the reference interface, a mesh-displacement field $\widehat{\mathbf{w}}$ is constructed on Ω_f^0 by solving a pseudo-elastic extension problem (cf. Section 8.2) with Dirichlet data $\widehat{\mathbf{w}} = \mathbf{d}$ on Γ_{if}^0 and $\widehat{\mathbf{w}} = \mathbf{0}$ on rigid boundaries, and $\Omega_f = \{X + \widehat{\mathbf{w}}(X) : X \in \Omega_f^0\}$. Because time does not appear in the baseline problem, no mesh velocity enters the fluid equations and no geometric-conservation-law constraint arises; such constraints become mandatory only for the transient extension, where the ALE formulation must be made GCL-compliant [13].

3.10 Constitutive completeness

No strain-gradient, nonlocal, fractional-derivative, memory, or two-temperature terms appear in this model, by deliberate restriction (cf. Section 11.1). Any such addition would require its own

constitutive evidence, boundary conditions, and validation objects, none of which are available in the literature base used here.

Chapter 4

Derivation of the plane-strain thermoelastic formulation

The relations used for design estimates and for the solid-solver sanity checks are derived here from the constitutive law (3.5). Where a simpler expression circulating in the working notes during preparation was found to be in error, the corrected expression is derived and the correction is indexed in the discrepancy register (Appendix B, Table B.1). The underlying constitutive theory is classical [8, 17]; the specific reductions for the present geometry are derived here and are not attributed to the literature.

4.1 Plane-strain reduction of the constitutive law

Write (3.5) componentwise with $e = \operatorname{tr}\boldsymbol{\varepsilon} = \varepsilon_{xx} + \varepsilon_{yy}$ (plane strain). In compliance form, $\varepsilon_{zz} = [\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})]/E + \alpha\theta = 0$ gives (3.6) directly. In stiffness form, $\sigma_{zz} = \lambda e - \beta\theta$; substitution of $\lambda = E\nu/[(1+\nu)(1-2\nu)]$, $\mu_L = E/[2(1+\nu)]$, and the in-plane stress–strain relations confirms that the two expressions coincide identically (algebra in Appendix A, cell A2).

For a strip with traction-free faces ($\sigma_{yy} = 0$ locally at each x), elimination of ε_{yy} through $\sigma_{yy} = 2\mu_L\varepsilon_{yy} + \lambda e - \beta\theta = 0$ yields $e = (2\mu_L\varepsilon_{xx} + \beta\theta)/(\lambda + 2\mu_L)$; back-substitution into $\sigma_{xx} = 2\mu_L\varepsilon_{xx} + \lambda e - \beta\theta$, using $\lambda + 2\mu_L = E(1-\nu)/[(1+\nu)(1-2\nu)]$, produces the one-dimensional wide-beam law

$$\sigma_{xx} = \frac{E}{1-\nu^2}\Big[\varepsilon_{xx} - (1+\nu)\,\alpha\theta\Big]. \tag{4.1}$$

Form (4.1) makes effective constants $E/(1-\nu^2)$ and $(1+\nu)\alpha$ explicit; both appear in the curvature and buckling relations below and distinguish the plane-strain strip from a narrow beam [8]. Dimensions: E [Pa] multiplies a dimensionless strain; $(1+\nu)\alpha\theta$ is dimensionless.

4.2 Mean-stress identity

From (3.8) and (3.6), $\sigma_m = (\sigma_{xx} + \sigma_{yy} + \sigma_{zz})/3$ takes the stress form

$$\sigma_m = \frac{1+\nu}{3}\big(\sigma_{xx} + \sigma_{yy}\big) - \frac{E\alpha}{3}\,\theta. \tag{3.8a}$$

Summing the in-plane constitutive relations gives $\sigma_{xx} + \sigma_{yy} = 2(\lambda + \mu_L)e - 2\beta\theta$; inserting this in (3.8a) and using $2(1+\nu)(\lambda+\mu_L) = E/(1-2\nu) = 3K$ and $2(1+\nu)\beta + E\alpha = 3\beta$ (Appendix A) yields the strain form

$$\sigma_m = K\big(e - 3\alpha\theta\big), \tag{3.8b}$$

i.e. mean stress equals bulk modulus × mechanical volumetric strain. The two forms are algebraically identical and serve as mutual checks in post-processing: the first requires complete Cauchy components (never the mechanical part alone), the second requires only $\operatorname{tr}\boldsymbol{\varepsilon}$ and θ.

4.3 Reference equilibrium states (sanity checks)

Three elementary states are used to test the constitutive implementation. **(a) Fully constrained body, uniform rise θ ($\boldsymbol{\varepsilon} \equiv 0$).** Then $\boldsymbol{\sigma} = -\beta\theta\mathbf{I}$, so $\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = -\beta\theta$ and

$$\sigma_m = -3K\alpha\theta = -\beta\theta. \tag{4.2}$$

(b) Axially restrained plane-strain strip with traction-free faces, uniform θ. Constraints $\varepsilon_{xx} = 0$, $\sigma_{yy} = 0$, $\varepsilon_{zz} = 0$ give, from (4.1) and (3.6),

$$\sigma_{xx} = \sigma_{zz} = -\frac{E\alpha\theta}{1-\nu}, \qquad \varepsilon_{yy} = \frac{1+\nu}{1-\nu}\,\alpha\theta, \qquad \sigma_m = -\frac{2E\alpha\theta}{3(1-\nu)}. \tag{4.3}$$

The mean stress of case (b), $-2E\alpha\theta/[3(1-\nu)]$, differs from $-3K\alpha\theta$: the two values belong to different restraint conditions and were conflated in an earlier draft (Appendix B, register D-12); both are retained, correctly paired, as unit-test expectations in Chapter 9. **(c) Free plane-strain strip, uniform θ.** With $\sigma_{xx} = \sigma_{yy} = 0$, equation (3.6) forces $\sigma_{zz} = -E\alpha\theta$, and

$$\sigma_m = -\frac{E\alpha\theta}{3} \neq 0. \tag{4.4}$$

Consistency with (3.8b) follows from $e = \varepsilon_{xx} + \varepsilon_{yy} = 2(1+\nu)\alpha\theta$, which gives $K\,[2(1+\nu) - 3]\alpha\theta = E(2\nu - 1)\alpha\theta/[3(1-2\nu)] = -E\alpha\theta/3$. This value is the plane-strain analogue of the stress-free three-dimensional expansion and is the correct expectation for the strip test (register D-2).

4.4 Free thermal curvature of the strip

Let the through-thickness temperature be linear, $\theta(y) = \bar{\theta} + \Gamma y$ measured from the mid-plane with $\Gamma = \Delta T_t/d$, and the strip be free of external loads with traction-free faces. With $\varepsilon_{xx} = \varepsilon_0 + \kappa y$ in (4.1), zero axial force resultant gives $\varepsilon_0 = (1+\nu)\alpha\bar{\theta}$, and zero moment resultant, $\int_{-d/2}^{d/2} \sigma_{xx}\, y\, \mathrm{d}y = 0$, gives the free thermal eigen-curvature

$$\kappa_{\mathrm{th}} = (1+\nu)\,\alpha\,\frac{\Delta T_t}{d} = (1+\nu)\,\alpha\,\frac{q''}{k_s}, \tag{4.5}$$

where the second form follows from one-dimensional steady conduction, $\Delta T_t = q'' d/k_s$ [11]; (4.5) is the plane-strain specialisation of the classical thermal-curvature result [8]. At $\nu = 0$ the narrow-beam value $\kappa = \alpha\Delta T_t/d$ is recovered; a factor $(1+\nu)/(1-\nu)$ would pertain to biaxially constrained curvature, not to the present free bending, and its appearance in an earlier draft is recorded in register D-1. Two consequences matter for the design. First, κ_{th} is independent of d: the through-wall gradient adjusts exactly to cancel the thickness. Second, a clamped–clamped strip at uniform κ_{th} carries a constant corrective moment $M_0 = D_s\kappa_{\mathrm{th}}$ and does not bow at all ($w \equiv 0$); pinned supports, or axial non-uniformity of $\kappa_{\mathrm{th}}(x)$, are required for the bowing mechanism to exist, which fixes the support choice of Assumption A9.

4.5 Pinned–pinned thermal bowing

For the reduced wall equation (3.9) with no mechanical load, the curvature relation is $w''(x) = -\kappa_{\mathrm{th}}(x)$, with w positive into the channel (Section 2.3). For constant κ_{th} and $w(0) = w(L) = 0$, integration gives

$$w(x) = \frac{\kappa_{\mathrm{th}}}{2}\,x(L-x), \qquad w_{\max} = \frac{\kappa_{\mathrm{th}}L^2}{8}. \tag{4.6}$$

Dimensions: $\mathrm{m}^{-1} \times \mathrm{m}^2 = \mathrm{m}$. Relation (4.6) is exact within the thin-plate framework *at negligible axial force*, which is the state of the free-ended strip and of the zero-membrane benchmark specimen of Chapter 9 ($L/d = 25$, keeping shear-deformation corrections below 1% [16]). For the axially constrained strip the thermal membrane force of Section 4.6 is not negligible at the baseline, and the bowing relation must be replaced by the beam–column form of Section 4.7.

4.6 Thermal membrane force and pinned–pinned buckling

With the axial displacement blocked at both ends (A9), the mean axial strain vanishes and integration of (4.1) over the thickness yields the compressive membrane force per unit depth

$$N_\theta = -\frac{Ed\,\alpha\bar{\theta}}{1-\nu}. \tag{4.7}$$

A sign convention is fixed here and used throughout: N_θ is the *signed* axial resultant (tension positive, so thermal compression gives $N_\theta < 0$), while the beam–column parameter and every force ratio use the *positive compressive magnitude* $N_\theta^c = |N_\theta|$, so that $N_\theta^c/N_{\mathrm{cr}} \in (0,1)$ is sub-critical and $\mu^2 = N_\theta^c/D_s > 0$. The compatibility statement of Section 4.7 acts on the signed resultant; its correction term is positive and therefore reduces the compression. The pinned–pinned Euler threshold for the strip, using the wide-plate flexural rigidity of (3.9), is $N_{\mathrm{cr}} = \pi^2 D_s/L^2$ [16]; the wide-plate value is the consistent one here *because plane strain suppresses the antielastic (out-of-plane) curvature mode*, so the strip cannot shed bending stiffness into the out-of-plane direction — a strip free to anticlastically buckle would carry the reduced rigidity $(1-\nu^2)\,Ed^3/12$. Equating $|N_\theta| = N_{\mathrm{cr}}$ gives the critical mean temperature rise

$$\bar{T}_{\mathrm{cr}} = \frac{\pi^2 d^2}{12\,\alpha(1+\nu)L^2}. \tag{4.8}$$

The clamped–clamped eigenvalue $4\pi^2 D_s/L^2$ would give a threshold four times higher; the pinned value (4.8) is the one consistent with the adopted supports and replaces the clamped-based estimate used in an early design note (Appendix B, register D-4). The admissibility rule is $\bar{\theta}_{\max} \le \bar{T}_{\mathrm{cr}}/M$, with margin $M = 2$ used in Chapter 7. The full-edge pin idealization of Assumption A9 stiffens the ends by $O(d/L)$ relative to ideal point pins; the catalogue is therefore read as accurate to a few per cent at the baseline aspect ratio, with the one-time numerical check of Chapter 9 closing the residual.

4.7 Beam–column bowing at finite membrane force

The linear bowing of Section 4.5 and the membrane force of Section 4.6 are not independent: a pinned strip bearing a compressive force $N_\theta^c = |N_\theta|$ couples its transverse deflection to that force at leading order, which is precisely the mechanism behind the threshold (4.8). The small-strain, moderate-rotation (von Kármán) wall model for this strip consists of the transverse equilibrium equation *together with* an axial compatibility condition that links the membrane force to the deflection. What is adopted here is the *reduced (frozen-force) beam–column equation*: the transverse member alone, retaining the full geometric-stiffness contribution of the compressive membrane force but freezing that force at its blocked-state value N_θ^c:

$$D_s\,w''''(x) + N_\theta^c\,w''(x) = 0, \qquad w(0) = w(L) = 0, \qquad w''(0) = w''(L) = -\kappa_{\mathrm{th}}, \tag{4.9}$$

where the end condition on w'' states that the *total* bending moment $M = D_s\,(w'' + \kappa_{\mathrm{th}})$ vanishes at the pins. With $\mu^2 = N_\theta^c/D_s$ the exact solution is

$$w(x) = \frac{\kappa_{\mathrm{th}}}{\mu^2}\left[\frac{1-\cos\mu L}{\sin\mu L}\,\sin\mu x + \cos\mu x - 1\right], \tag{4.10}$$

which recovers the parabola (4.6) as $\mu L \to 0$ and diverges at $\mu L = \pi$, i.e. exactly at the Euler threshold (4.8); the buckling limit and the deflection law are therefore two limits of one relation, as they must be. The fully consistent von Kármán formulation would additionally update the force through axial compatibility: with both end edges blocked, $\langle u' \rangle = 0$ in the mean strain $\langle \varepsilon_x \rangle = \langle u' \rangle + \tfrac{1}{2}\langle w'^2 \rangle - (1+\nu)\alpha\bar{\theta}$, so the signed resultant acquires the arc-stretching correction

$$N_\theta(w) \;=\; N_\theta \;+\; \frac{Ed}{2(1-\nu^2)}\,\langle w'^2 \rangle, \tag{4.11}$$

and the deflection problem becomes nonlinear. Equation (4.9) is the first iterate of that system — the force frozen at $N_\theta^c = |N_\theta|$ — and its accuracy against the consistent solution is quantified below rather than assumed. At the baseline parameters ($N_\theta^c/N_{\rm cr} = 0.275$, $\mu L = 1.65$):

$$w_{\max} = 2.19{\times}10^{-4}\,\mathrm{m} = 0.438\,H_0 \quad \text{(frozen force, adopted)}, \qquad \frac{w_{\max}^{\rm lin}}{H_0} = 0.315 \quad \text{(linear, Eq. (4.6))}, \tag{4.12}$$

an amplification by the factor 1.39, consistent at leading order with the beam–column factor $1/(1{-}N_\theta^c/N_{\rm cr}) = 1.38$. The maximum slope is $w'_{\max} = 0.027$ rad, the bending fibre strain is $(\kappa_{\rm geo}{-}\kappa_{\rm th})d/2 = 9.9\times10^{-4}$ with $\kappa_{\rm geo} = -w''$, and the maximum total mechanical surface strain is $\approx 5.4{\times}10^{-3}$; the von Kármán hypotheses (strains $\ll 1$, rotations $\ll 1$ rad) are comfortably satisfied, so (4.9) is an appropriate model level — not a large-deformation extension, but a reduction of the von Kármán system whose error must be stated. Freezing the force is an approximation, not an identity: the deflection relaxes the blocked-end mechanical strain $e_0 = (1{+}\nu)\alpha\bar\theta = 2.26{\times}10^{-3}$ by the arc-stretching term $\frac{1}{2}\langle w'^2\rangle = 1.8\times10^{-4}$ (circular-arc estimate 1.6×10^{-4}), i.e. a first-iterate force relaxation of 6.1%. Solving the compatibility relation (4.11) self-consistently (a scalar fixed point, since the solution family (4.10) holds for every fixed force) gives a relaxation of 5.9% and lowers the baseline deflection to $w_{\max} = 2.14\times10^{-4}\,\mathrm{m} = 0.428\,H_0$: the frozen-force value (4.12) *over*predicts the consistent deflection by 2.3%, a one-sided (conservative) error that never changes sign. Within the admissible envelope the error remains below $\approx 4\%$ (the deformation and stress ceilings of Chapter 7 bind at $N_\theta^c/N_{\rm cr} = 0.30$ and 0.33), the mid-span moment and bending stress shift by under 8% in the same safe direction, and the buckling margin is exact — the relaxation term vanishes as the deflection vanishes, so the onset threshold is governed by the blocked force and $\bar T_{\rm cr}$ is unaffected. The error grows rapidly only beyond $N_\theta^c/N_{\rm cr} \approx 0.5$ (it reaches $\approx 17\%$ at the $M_b = 2$ ceiling), which is precisely where the escalation criterion of Chapter 8 moves the computation off the frozen-force form.

Three consequences fix the model specification. First, the solid model of the coupled problem is specified at the reduced (frozen-force) von Kármán level — the geometric-stiffness contribution of the prescribed membrane state, of which the linear model is the small-force limit (at $N_\theta^c/N_{\rm cr} \lesssim 0.1$ the two differ by $\lesssim 10\%$) — while the implementation re-evaluates the membrane state across coupled iterations (Chapter 8), so the frozen-force form applies within each solve and the consistency update (4.11) is recovered at the fixed point. Second, the earlier design relation (4.6) is retained for the free and zero-membrane states only, and the baseline bowing is quoted from (4.12). Third, the bending stress follows from the beam–column moment: at mid-span $M = -6.4$ N m per unit depth, giving $\sigma_{\rm bend} = 6|M|/d^2 = 4.2$ MPa (the unreleased-moment scale $E\kappa_{\rm th}d/2/(1-\nu^2) = 9.0$ MPa quoted previously belongs to a fully restrained-curvature state and does not occur in the pinned strip; Appendix B, register D-17).

4.8 Hydraulic loading and the deformation ratio

Pressure drop. Integrating the plane-Poiseuille profile $u(y) = 6U(y/H_0)(1 - y/H_0)$ [2, 3] through the momentum balance, $\mathrm{d}p/\mathrm{d}x = \mu u'' = -12\mu U/H_0^2$, gives the slit pressure drop

$$\Delta p = \frac{12\mu U L}{H_0^2} \qquad \Big[\frac{(\mathrm{Pa\,s})(\mathrm{m/s})(\mathrm{m})}{\mathrm{m}^2} = \mathrm{Pa}\Big]. \tag{4.13}$$

A predecessor note carried the same expression with H_0^3 in the denominator, which is dimensionally not a pressure (Pa/m) and produced an estimate three orders of magnitude too large; the corrected value is computed in Chapter 7 (register D-5).

Pressure deflection. For a *pinned–pinned* strip under uniform transverse load $\bar p$ opposing the bowing, the standard bending solution [16, 17] gives

$$w_p = \frac{5\,\bar p L^4}{384 D_s}. \tag{4.14}$$

(The coefficient 5/384 belongs to pinned supports; the value 1/384 used in an earlier estimate is the *clamped–clamped* coefficient. See register D-13.) The load level requires a datum statement, because the incompressible pressure *level* on the wall is fixed by the outlet datum of condition B2 and by the casing environment of Assumption A9, not by Δp alone: with the datum $p = 0$ at the outlet and the casing at the same reference pressure, the gauge pressure on the strip rises linearly from $p(L) = 48$ Pa to $p(0) = 120$ Pa (gradient $12\mu U/H_0^2$ over the strip and the exit length), so the strip carries a linearly varying load with mean

$$\bar{p} = \tfrac{1}{2}\,[p(0) + p(L)] = 84 \text{ Pa} \tag{4.15}$$

rather than the value $\Delta p/2 = 36$ Pa quoted in earlier drafts (register D-16); the uniform-load formula (4.14) is then read as a design estimate for this mildly varying load. With $\bar{p}$ of (4.15), $w_p = 9.2 \times 10^{-8}$ m, and with w_{th} of (4.12), the deformation ratio

$$\Gamma_p = \frac{w_{\mathrm{th}}}{w_p} = \frac{48\,\kappa_{\mathrm{th}} D_s}{5\,\bar{p} L^2} = 1.7 \times 10^3 \quad \text{at the baseline,} \tag{4.16}$$

measures how strongly the deformation is thermally, rather than hydraulically, driven; the conclusion (thermal dominance, $\Gamma_p \gg 5$) is insensitive to the datum choice, since any plausible casing pressure shifts $\bar{p}$ by less than an order of magnitude. Because κ_{th} is independent of d while $D_s \propto d^3$ and $\bar{p} \propto U H_0^{-2}$, the ratio grows with thicker walls and decreases with speed and with reduced height: these are the control directions examined at the design stage.

4.9 Time scales and the steady-state reduction

Four time scales govern the start-up problem (inputs of Table 2.2; arithmetic in Appendix A): the structural wave crossing $t_{\mathrm{wave}} = d\sqrt{\rho_s/E} = 1.8 \times 10^{-6}$ s; the fluid advection time $t_{\mathrm{adv}} = L/U = 0.6$ s; the fluid transverse diffusion time $t_f = H_0^2/\kappa_f \approx 1.8$ s; and the solid conduction time $t_s = d^2/\kappa_s \approx 51$ s. The ordering $t_{\mathrm{wave}} \ll \{t_{\mathrm{adv}}, t_f\} \lesssim t_s$ has three consequences. First, mechanical equilibrium is quasi-static on all observable transport times, justifying (3.7). Second, the steady coupled solution targeted by the baseline is the $t \to \infty$ asymptote of the start-up dynamics, which is rate-limited by the solid conduction time, not by the fluid. Third, in a future start-up computation the fluid transients may be neglected at leading order ($t_f/t_s \approx 0.04$) provided an a-posteriori check is made; the thermoelastic deformation-heating source of Section 3 vanishes identically in the steady problem, and its transient neglect is controlled by $\xi \approx 2 \times 10^{-2}$ [8].

Chapter 5

Boundary and interface conditions

5.1 Interface conditions on the fluid–solid boundary

On the deformed interface Γ_{if} the two subsystems exchange kinematics, tractions, and heat. With the conventions of Section 2.3 the conditions are

$$\mathbf{u}\big(X + \mathbf{d}(X)\big) = 0 \qquad \text{no-slip condition,} \tag{5.1}$$

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \big(- p\,\mathbf{I} + 2\mu \mathbf{D}(\mathbf{u})\big) \cdot \mathbf{n} \qquad \text{traction continuity,} \tag{5.2}$$

$$T_f = T_s \qquad \text{temperature continuity,} \tag{5.3}$$

$$-k_f \nabla T_f \cdot \mathbf{n} = -k_s \nabla T_s \cdot \mathbf{n} \qquad \text{heat-flux continuity.} \tag{5.4}$$

The kinematic condition (5.1) is the steady form of the usual $\mathbf{u} = \partial \mathbf{d}/\partial t$; the traction condition (5.2) follows from balance of the interface traction vector in the absence of interfacial stress, $\boldsymbol{\sigma}^{(s)} \cdot \mathbf{n} + \boldsymbol{\sigma}^{(f)} \cdot \mathbf{n}_f = 0$ with $\mathbf{n}_f = -\mathbf{n}$, where $\boldsymbol{\sigma}^{(f)} = -p\mathbf{I} + 2\mu\mathbf{D}$ is the fluid stress. Evaluating both sides at the same point and with the single normal $\mathbf{n}$ yields (5.2); the viscous part of the traction is retained alongside the pressure, although at the baseline the pressure part dominates by about two orders of magnitude in magnitude of its bending effect ($\tau_w = 0.6$ Pa against $\bar{p} = 84$ Pa; Chapter 7). The conditions (5.3)–(5.4) express perfect thermal contact (A8): continuity of temperature (no contact resistance) and continuity of the normal heat flux, with the flux vector $\mathbf{q} = -k\nabla T$ evaluated with a single normal, following the standard conjugate-heat-transfer formulation [11].

5.2 Exterior boundary conditions

Table 5.1 specifies the remaining boundary data. The inlet profile is the fully developed plane-Poiseuille parabola for a rigid entry of equal height, so that the mass flow per unit depth is fixed at $\dot{m}' = \rho U H_0$ at both fixed-U operation and (for the future fixed-Δp variant) through the equivalent boundary data; the entrance length is a design input.

Table 5.1. Boundary-condition specification.

ID	Boundary	Condition	Remark
B1	$x = -L_e$, fluid	$\mathbf{u} = 6U\,\frac{y}{H_0}\big(1 - \frac{y}{H_0}\big)\hat{\mathbf{x}}$, $T_f = T_{\text{in}}$	fixed mass flow, rigid entry
B2	$x = L + L_o$, fluid	traction-free outflow with datum $p = 0$; $\partial_n T_f = 0$	exit-length/backflow sensitivity required (App. B, C-9)
B3	lower wall $y = 0$	$\mathbf{u} = 0$, $\partial_n T_f = 0$	rigid, adiabatic

ID	Boundary	Condition	Remark
B4	rigid upper sections	$\mathbf{u} = 0$, $\partial_n T_f = 0$	rigid casing
B5	Γ_{if}	(5.1)–(5.4)	conjugate interface
B6	outer face Γ_h^0	$-k_s \nabla T_s \cdot \mathbf{n}_{\text{out}} = q''$	baseline regime N
B7	strip ends $X = 0, L$	$d_1 = d_2 = 0$, zero applied moment; $\partial_n T_s = 0$	pinned, adiabatic (A9)
B8	remaining solid faces	zero traction, zero flux	insulated casing
B9	variant of B6	$T_s = T_h$ on Γ_h^0	regime D; matched heat-input rule (A10)

Two conditions deserve comment. The outflow condition B2 is the usual "do-nothing" natural condition of finite-element practice for (3.2); it requires the exit length L_o to be sufficient that the flow has re-developed a nearly parallel state, verified a posteriori, and no backflow occurs. The datum $p = 0$ in B2, together with the casing reference of Assumption A9, fixes the gauge pressure carried by the deformable strip (Section 4.8); this datum is part of the problem specification and must be reported with every computed case. The outer-face condition B6 fixes the areal heat input and thereby the total heat duty per unit depth, $Q' = q'' L$, which is the quantity balanced in the global enthalpy check of Section 8.4; regime D (B9) cannot share the heat-duty bookkeeping with regime N and is compared only at matched Q'.

5.3 Interface data flow in the coupled problem

With the thermal load imposed on the solid side, the data flow of the coupled problem is: the fluid subproblem receives its domain and the interface temperature from the solid; the solid subproblem receives the interface traction (pressure-dominated) and the conjugate flux condition. Because the solid problem is linear, the displacement decomposes exactly as

$$\mathbf{d} = \mathbf{d}_{\text{th}} + \mathbf{d}_p, \tag{5.5}$$

where $\mathbf{d}_{\text{th}}$ solves the thermoelastic problem with zero interface traction and $\mathbf{d}_p$ solves the isothermal problem under the fluid traction. This superposition is an algebraic identity of the *linear* solid model at fixed configuration and fixed loads; it is used as the mechanism-attribution device between thermally driven and traction-driven deformation. Two scope limits are recorded. First, across the coupled iterations the fluid state and the traction distribution depend on $\mathbf{d}$, so the decomposition is exact only at the converged state and for the loads of that state. Second, at the von Kármán model level of Section 4.7 the geometric-stiffness term couples the two load systems, so the additive identity becomes approximate, with a relative error controlled by $N_\theta^c / N_{\text{cr}}$ ($\approx 28\%$ in $w_{\max}$ at the baseline, from the beam–column amplification); the implementation therefore attributes mechanisms by *separate load-case solutions* (thermal only, traction only, both) rather than by adding fields, and verifies exact additivity on a detuned case at $N_\theta^c / N_{\text{cr}} \leq 0.05$ (Chapter 9).

Chapter 6

Nondimensionalization and characteristic parameters

The governing groups are obtained by substituting the scales of the problem into the field equations of Chapter 3: length H_0, velocity U, pressure $\mu U/H_0$ (with the equivalent inertial statement noted where used), temperature differences $\Delta T^* = q'' H_0/k_f$, stress $E\alpha\Delta T^*$, displacement $\alpha\Delta T^* H_0$. No group is introduced independently of the equations.

Length-basis and friction-factor convention. Two Reynolds numbers appear in the channel literature: $\mathrm{Re}_{H_0} = \rho U H_0/\mu$ and $\mathrm{Re}_{D_h} = \rho U(2H_0)/\mu = 2\mathrm{Re}_{H_0}$, with $D_h = 2H_0$ for the slit. The Darcy friction factor is defined by $\Delta p = f_D\,(L/D_h)\,\rho U^2/2$, and for the slit $f_D\,\mathrm{Re}_{D_h} = 96$ [2]; the numerator differs by a factor four from the Fanning convention ($f_F\,\mathrm{Re}_{D_h} = 24$). Throughout this document the D_h basis and the Darcy convention are used, and the basis is required to be printed on every computed output (Appendix B, register D-6).

6.1 Groups from the fluid equations

Substituting the scales into (3.2): the inertia-to-viscous ratio

$$\mathrm{Re}_{D_h} = \frac{\rho U(2H_0)}{\mu} \quad (= 50 \text{ at baseline}), \tag{6.1}$$

and from (3.3) the advection-to-conduction ratio

$$\mathrm{Pr} = \frac{\mu c_p}{k_f} = 6.97, \qquad \mathrm{Pe} = \mathrm{Re}_{D_h}\mathrm{Pr} = 348. \tag{6.2}$$

Were the Boussinesq term retained, the ratio of buoyant to inertial forcing would be measured by the Grashof and Richardson numbers

$$\mathrm{Gr} = \frac{g\beta_f \Delta T^* H_0^3}{\nu_{\mathrm{kin}}^2}, \qquad \mathrm{Ri} = \frac{\mathrm{Gr}}{\mathrm{Re}_{H_0}^2} = \frac{g\beta_f \Delta T^* H_0}{U^2}, \tag{6.3}$$

where the second form makes the length basis explicit: both numerator and denominator of Ri must carry the *same* length, so with the Grashof number built on H_0 and the temperature scale $\Delta T^* = q'' H_0/k_f$ (itself fixed by H_0), the consistent Reynolds number is $\mathrm{Re}_{H_0} = \rho U H_0/\mu$, not Re_{D_h}; equivalently, on the D_h basis both groups change ($\mathrm{Gr}_{D_h} = 8\,\mathrm{Gr}_{H_0}$) and Ri changes by the factor $D_h/H_0 = 2$. The two conventions differ by a factor of two and both are quoted in Chapter 7. Finally, the dissipation group

$$\mathrm{Br} = \frac{\mu U^2}{k_f \Delta T^*} \tag{6.4}$$

controls Assumption A7. All four are evaluated in full arithmetic detail in Section 7.1; an earlier draft divided $\mathrm{Gr}(H_0)$ by $\mathrm{Re}_{D_h}^2$ and mis-transcribed the powers of ten, values now corrected (Appendix B, register D-14).

6.2 Groups from the solid and interface equations

Continuity of the interface flux with the common scale ΔT^* brings in the conductivity ratio and the two geometric ratios

$$R_k = \frac{k_s}{k_f} = 0.42, \qquad D_r = \frac{d}{H_0} = 6, \qquad \Lambda = \frac{L}{H_0} = 60, \tag{6.5}$$

written R_k and D_r to avoid collision with the bulk modulus K and the rate-of-deformation tensor $\mathbf{D}$ (Appendix B, register D-18). The through-wall temperature drop expressed in units of ΔT^* is $\Delta T_t/\Delta T^* = D_r/R_k = 14.4$: essentially the entire thermal gradient of the coupled problem lives inside the wall, which is the quantitative basis of Assumption A2's self-consistency check and the reason the solid sees a much larger θ than the fluid ever will.

The characteristic result of the model is that the relative thermal bowing of the channel, from (4.5), (4.6), and $\Delta T_t = q'' d/k_s$, collapses to

$$\frac{w_{\max}^{\mathrm{lin}}}{H_0} = \frac{(1+\nu)\,\Sigma_\kappa\,\Lambda^2}{8}, \qquad \text{with} \qquad \Sigma_\kappa = \alpha\,\frac{q'' H_0}{k_s} \tag{6.6}$$

a dimensionless curvature-strain number. At baseline $(1+\nu)\Sigma_\kappa = 7\times10^{-4}$ and $w_{\max}^{\mathrm{lin}}/H_0 = 0.315$; at the baseline membrane force the frozen-force beam–column form of Section 4.7 gives the adopted $w_{\max}/H_0 = 0.438$ (self-consistent comparison 0.428), and (6.6) is read as the linear limit of that relation. Equation (6.6) assembles geometry, material, and load into a single ordinate and is the ordering parameter of the regime analysis in Chapter 7 and of any future parameter study.

The ratios governing the mechanical regime are the deformation ratio of Section 4.8, $\Gamma_p = 48\kappa_{\mathrm{th}} D_s/(5\bar{p}L^2)$, and the buckling sub-criticality margin

$$M_b = \frac{\bar{T}_{\mathrm{cr}}}{\bar{\theta}} \qquad (\text{required } M_b \geq 2), \tag{6.7}$$

with $\bar{T}_{\mathrm{cr}}$ of (4.8) and $\bar{\theta}$ the mean wall temperature rise.

6.3 Admissibility bounds on the regime

Any reported operating point of the model must lie inside the following bounds, all verified arithmetic at the baseline in Chapter 7:

- A-1 $\mathrm{Re}_{D_h} \leq 100$ (laminar-design headroom, A6);
- A-2 $M_b \geq 2$ (sub-critical pinned supports, Section 4.6);
- A-3 kinematics bounds (von Kármán validity, A4 and Section 4.7): bending fibre strain $(\kappa_{\mathrm{geo}} - \kappa_{\mathrm{th}})d/2 \leq 5\times10^{-3}$, total mechanical surface strain $\leq 10^{-2}$, arc-stretching strain $\leq 10^{-3}$, maximum slope ≤ 0.1 rad, plus the mesh-generation bound $|w|/H_0 \leq 0.5$ with positive element Jacobians;
- A-4 fluid-property drift below $\sim 10\%$ over the computed fluid-side temperature range (A2);
- A-5 $\Gamma_p \geq 5$ for a case to be classified as thermally driven;
- A-6 computed wall stress $\sigma_{\max} \leq 0.2\,\sigma_y$, with σ_y from the frozen datasheet band (design cap with margin ≥ 5; register D-19 restates the earlier ambiguous wording).

6.4 Dependence on yet-unfrozen data

Only temperature differences appear in the equations, so T_{ref} and T_{in} enter the solution exclusively through property values; both are to be fixed at the property-data freeze (Appendix B, C-6), as is the value of β_f entering Gr. No dimensionless number above depends on an invented constant: groups whose inputs are still datasheet-conditional are listed with that dependency in Table 7.1.

Chapter 7

Baseline operating conditions and analytical calculations

Every quantity below is recomputed from the inputs of Tables 2.1 and 2.2 by the equations of Chapters 3–6; the arithmetic cells in Appendix A show representative substitutions. Quantities that depend on datasheet values not yet frozen (Appendix B, C-6) carry the remark "datasheet dependent" and would shift with the frozen data, without structural change to the analysis.

7.1 Baseline calculation table

Table 7.1. Analytical baseline at the design point. "R" refers to the equation number where the expression is derived. Datasheet-dependent inputs: see Table 2.2.

Quantity	Expression	Substitution and result	Basis
Re_{D_h}	$2\rho U H_0/\mu$	$2(10^3)(5 \times 10^{-2})(5 \times 10^{-4})/10^{-3} = 50$	(3.2)
Pr; Pe	$\mu c_p/k_f$; $\mathrm{Re}_{D_h}\mathrm{Pr}$	6.97; 348	(3.3)
f_D (rigid check)	$96/\mathrm{Re}_{D_h}$	1.92	[2]
Δp	$12\mu UL/H_0^2$, (4.13)	72 Pa; checks: $f_D(L/D_h)\rho U^2/2 = 72$ Pa; $\Delta p H_0 = 2\tau_w L$ with $\tau_w = 0.6$ Pa	(4.13)
ΔT^*	$q'' H_0/k_f$	$(5\times10^3)(5\times10^{-4})/0.6 =$ 4.17 K	scale of Ch. 6
ΔT_t	$q'' d/k_s$	$(5 \times 10^3)(3 \times 10^{-3})/0.25 = 60$ K	[11]; datasheet dependent
κ_{th}	$(1+\nu)\alpha q''/k_s$, (4.5)	$1.4(5 \times 10^{-5})(5 \times 10^3)/0.25 = 1.4$ m^{-1}	(4.5); datasheet dependent
$w_{\max}$ (linear)	$\kappa_{\mathrm{th}} L^2/8$, (4.6)	$1.4(9 \times 10^{-4})/8 = 1.58 \times 10^{-4}$ m	(4.6)
$w_{\max}$ (beam–column, frozen force)	(4.10), $\mu L = 1.65$	2.19×10^{-4} m; $w_{\max}/H_0 = 0.438$ (self-consistent comparison: 2.14×10^{-4} m, $0.428\,H_0$)	(4.12)

Quantity	Expression	Substitution and result	Basis
$w_{\max}/H_0$	(6.6); (4.12)	0.315 (linear); 0.438 (beam–column); A-3 check below	(6.6)
fibre / arc strain	$(\kappa_{\mathrm{geo}} - \kappa_{\mathrm{th}})d/2$; $\frac{1}{2}\langle w'^2\rangle$	9.9×10^{-4}; 1.8×10^{-4}	(4.9); (4.11)
$\bar{\theta}$ (mean wall rise)	$\Delta T_t/2 + q''/h + \Delta T_b/2$	$30 + 1.6 + 0.7 \approx 32$ K	estimate; see text
N_θ	$-Ed\alpha\bar{\theta}/(1-\nu)$, (4.7)	$-(3.6\times10^9)(3\times10^{-3})(5\times10^{-5})(32)/0.6 = -2.9\times10^4$ N/m	(4.7)
D_s	$Ed^3/[12(1-\nu^2)]$	$(3.6\times10^9)(2.7\times10^{-8})/10.08 = 9.6$ N m	(3.9)
N_{cr}	$\pi^2 D_s/L^2$	$9.87(9.6)/(9\times10^{-4}) = 1.06\times10^5$ N/m	[16]
$\bar{T}_{\mathrm{cr}}$	(4.8)	$\pi^2(3\times10^{-3})^2/[12(5\times10^{-5})(1.4)(9\times10^{-4})] = 117$ K	(4.8)
M_b	$\bar{T}_{\mathrm{cr}}/\bar{\theta}$	$117.5/32.3 = 3.64$ (≥ 2, bound A-2 met)	def. Ch. 6
w_p	$5\bar{p}L^4/(384D_s)$, $\bar{p}$ from (4.15)	$5(84)(8.1\times10^{-7})/(384\cdot9.6) = 9.2\times10^{-8}$ m	(4.14)
Γ_p	$w_{\max}/w_p$, (4.16)	$2.19\times10^{-4}/9.2\times10^{-8} = 2.4\times10^3$ (≥ 5, A-5 met)	(4.16)
σ_{bend}	$6\lvert M_{\mathrm{mid}}\rvert/d^2$, M_{mid} from (4.9)	$6(6.4)/9\times10^{-6} = 4.2\times10^6$ Pa	(4.9)
σ_{memb}	$-E\alpha\bar{\theta}/(1-\nu)$	$-(3.6\times10^9)(5\times10^{-5})(32)/0.6 = -9.6\times10^6$ Pa	(4.1)
stress fraction	$\lvert\sigma\rvert_{\max}/\sigma_y$, $\lvert\sigma\rvert_{\max} \simeq 1.4\times10^7$ Pa	0.15 with $\sigma_y \simeq 9\times10^7$ Pa (A-6 cap $0.2\sigma_y$ met, datasheet band)	datasheet dependent
Gr	$g\beta_f\Delta T^* H_0^3/\nu_{\mathrm{kin}}^2$	$(9.81)(2.1\times10^{-4})(4.17)(1.25\times10^{-10})/10^{-12} = 1.07$	datasheet dependent (β_f)
Ri	$\mathrm{Gr}/\mathrm{Re}_{H_0}^2 = g\beta_f\Delta T^* H_0/U^2$, (6.3)	$1.07/625 = 1.7\times10^{-3}$ (3.4×10^{-3} on the D_h basis)	(6.3)
Br	$\mu U^2/(k_f\Delta T^*)$	$(10^{-3})(2.5\times10^{-3})/2.5 = 1.0\times10^{-6}$	(3.3) note
ΔT_b	$q''L/(\dot{m}'c_p)$, $\dot{m}' = \rho U H_0$	$150/(0.025\cdot4180) = 1.44$ K	energy balance
h_{rigid}; q''/h	$\mathrm{Nu}_{H1}k_f/D_h$	$(5.385)(0.6)/10^{-3} = 3231$ W m^{-2} K^{-1}; 1.55 K	[2] (H1 family)
t_{wave}	$d\sqrt{\rho_s/E}$	1.8×10^{-6} s	Sec. 4.9
t_s; t_f	d^2/κ_s; H_0^2/κ_f	51 s; 1.7 s	Sec. 4.9
ξ	$\beta^2 T_0/[(\lambda+2\mu_L)\rho_s c_s]$, $T_0 = 293$ K	2.2×10^{-2}	[8]
L_h; L_T (entry lengths, est.)	$0.05\,\mathrm{Re}_{D_h}D_h$; $0.05\,\mathrm{Pe}\,D_h$	2.5 mm; 17.4 mm	engineering estimates

The mean-wall-rise estimate $\bar{\theta}$ collects the one-dimensional through-wall mean drop $\Delta T_t/2$, the interface excess over the bulk fluid q''/h with the rigid-wall H1 coefficient, and half the bulk rise; it is an estimate with component-wise accuracy, and main sweeps will report $\bar{\theta}$ as a range rather than a point value (Appendix B, register D-7).

7.2 Interpretation

(i) What sets the deformation magnitude. Since $\kappa_{\mathrm{th}} = (1+\nu)\alpha q''/k_s$, the thermal curvature at fixed material scales linearly with the applied flux and is *independent of the wall thickness*: thickness cancels between the through-wall resistance and the curvature law. The relative bowing (6.6) is thus controlled by the geometric ratio $\Lambda = L/H_0$ and the material group α/k_s. At the baseline the linear relation gives $w^{\mathrm{lin}}_{\mathrm{max}}/H_0 = 0.315$ and the beam–column form of Section 4.7 — the frozen-force reduction appropriate to the strip's own membrane state — gives the adopted $w_{\mathrm{max}}/H_0 = 0.438$ (self-consistent comparison value 0.428; the reduction is conservative by 2.3%): the deformation is a substantial modification of the passage, and the kinematic conditions of bound A-3 (fibre strain 9.9×10^{-4}, total surface strain 5.4×10^{-3}, arc strain 1.8×10^{-4}, slope 0.027 rad) are met with wide margins.

(ii) Why this mechanism is specific to the wall material class. Repeating cell κ_{th} for stainless steel ($k_s = 16\ \mathrm{W\,m^{-1}\,K^{-1}}$, $\alpha = 1.2 \times 10^{-5}\ \mathrm{K^{-1}}$) at the same flux and geometry gives $\kappa_{\mathrm{th}} = 1.2 \times 10^{-5} \times 1.4 \times (5 \times 10^3)/16 = 5.3 \times 10^{-3}\ \mathrm{m^{-1}}$ and $w_{\mathrm{max}}/H_0 \approx 1.2 \times 10^{-3}$: a physically real deformation that is irrelevant for the convective field. The combination of low k_s and high α of the polymer class is what raises the bowing by more than two orders of magnitude; this material dependence, rather than any fluid-side property, selects the design space of the study.

(iii) The binding mechanical constraints are deformation and stress, with buckling as the outer ceiling. With the datum-consistent strip load, $\Gamma_p \approx 2.4 \times 10^3$ at the baseline: pressure-induced deflection is negligible against thermal bowing throughout the design space examined (bound A-5). Under the frozen-force beam–column relation the admissible heating is capped in sequence, all three ceilings scaling with $\kappa_{\mathrm{geo}} \sim \kappa_{\mathrm{th}}$ and $\bar{\theta} \propto q''$: the A-3 mesh-generation bound $|w|/H_0 \le 0.5$ is reached at $N^c_\theta/N_{\mathrm{cr}} = 0.30$, i.e. $q'' \approx 5.5 \times 10^3\ \mathrm{W\,m^{-2}}$; the wall-stress cap A-6 ($\sigma_{\mathrm{max}} \le 0.2\sigma_y$, $\sigma_{\mathrm{max}} \approx 13.9$ MPa at the baseline) at $N^c_\theta/N_{\mathrm{cr}} = 0.33$, i.e. $q'' \approx 5.9 \times 10^3\ \mathrm{W\,m^{-2}}$ ($w_{\mathrm{max}}/H_0 \approx 0.56$, $M_b = 3.1$, still sub-critical by A-2); and only at the extreme by buckling, $M_b \ge 2$ restricting $\bar{\theta} \le 58.8$ K, i.e. $q'' \approx 9.1 \times 10^3\ \mathrm{W\,m^{-2}}$ ($N^c_\theta/N_{\mathrm{cr}} = 0.5$, $w_{\mathrm{max}}/H_0 \approx 1.2$ frozen-force), at which the deflection would be of order H_0 and unacceptable in its own right. Within the envelope so bounded ($N^c_\theta/N_{\mathrm{cr}} \le 0.33$) the frozen-force reduction of Section 4.7 carries a deflection error below 4%, one-sided (conservative) against the self-consistent von Kármán solution. The design-ordering statement of the earlier draft — "the binding constraint is buckling" — is therefore corrected: deformation kinematics and stress bind first at the baseline geometry, and buckling caps the extreme (Appendix B, register D-15; the catalog of $\bar{T}_{\mathrm{cr}}$ versus thickness is in Appendix A, cell A4).

(iv) Stress state. The load-case decomposition (5.5) splits the wall stress into a bending part (beam–column extrema ± 4.2 MPa at the faces near mid-span) and a compressive membrane part (-9.7 MPa, uniform). The combined magnitude ~ 14 MPa is about 0.15 of the PEEK yield stress, inside the A-6 cap with margin ≥ 6 at the band edge; residual compressive membrane stress is the by-product of the axial constraint that also produces the buckling ceiling.

(v) Thermal bookkeeping. The fluid picks up only $\Delta T_b = 1.44$ K of bulk rise and an interface excess of about 1.6 K, while the wall sustains a 60 K through-thickness drop. Because the thermal entry length exceeds the heated length ($L/L_T \approx 1.7$), the developing-flow transfer coefficient along the strip is enhanced relative to the fully developed value used in the estimate of Section 7.1 (Table 7.1): the tabulated $\bar{\theta}$ is accordingly a conservative (upper) estimate, to be revised once computed fields exist. This partition (conductivity ratio $R_k = 0.42$, thickness ratio $D_r = 6$) confirms that constant fluid properties are self-consistent (bound A-4) and that the wall temperature field is dominated by the imposed load, not by feedback from the fluid

– the *deformation*, however, is exactly the first-order bookkeeping that changes the convective boundary condition.

7.3 Buoyancy disposition at the baseline

At the baseline Gr = 1.07 and, on the length-consistent definition (6.3), Ri = 1.7×10^{-3} (3.4×10^{-3} on the D_h basis; an earlier draft quoted Ri = 0.43 from a mixed-basis division with mis-transcribed powers of ten, corrected in register D-14). The buoyant forcing is therefore three orders of magnitude below the inertial forcing, and the constant-density equations (3.2) are adopted on the standard order-of-magnitude argument, without recourse to stratification stability. Two provisions are retained. First, a Boussinesq body-force term is kept as a dormant option in the weak form (one term), so that any future operating point with substantially larger fluid-side temperature excursions can be re-examined at negligible cost. Second, for context on the regime where the exclusion would fail — bulk Richardson numbers of order 10 and above, where stable stratification measurably modifies and deteriorates channel heat transfer — see [12]. The corrected estimate closes the question at the baseline; the disposition is revisited only if a parameter-study point drives Ri above $\mathcal{O}(10^{-2})$, which is recorded as a regime guard on the reported set.

Chapter 8

Proposed numerical formulation and solution procedure

This chapter specifies the discretization and solution procedure for the model of Chapters 3 and 5 at the level required for an independent implementation review. No code is presented and none is yet written (Appendix B); the specification is intended to be realisable with standard finite-element components in a general-purpose environment.

8.1 Spatial discretization

The fluid velocity and pressure are approximated by Taylor–Hood elements (continuous piecewise quadratic velocity, continuous piecewise linear pressure), satisfying the discrete inf–sup condition for the Stokes limit of (3.2) [1, 10]; both temperature fields and the solid displacement use the same quadratic family. The fluid energy equation (3.3) is strongly advection-dominated: with $\mathrm{Pe} = \mathrm{Re}_{D_h}\mathrm{Pr} = 348$ the cell Péclet number exceeds the Galerkin oscillation threshold on any mesh affordable at the required near-wall resolution, so the temperature transport is stabilized by a consistent streamline-upwind (SUPG) term with the standard residual-based coefficient; oscillation-free behaviour of the computed interface flux is part of the acceptance criteria of test V-B (Chapter 9). The meshes of the fluid and solid subdomains are constructed *conforming* at the interface: nodes and edges of the two meshes coincide on Γ_{if}. This choice is deliberate: interface quantities are then transferred by identical nodal values, the interface quadrature is exact to the geometric approximation order, and no mortar or projection operator – each of which would introduce its own consistency error in the conjugate flux condition – is required. Isoparametric quadratic boundary edges are used on the deformable boundary and are updated with the same map as the mesh displacement, so that the geometric approximation of Ω_f stays at the approximation order of the field variables.

8.2 Weak formulation and monolithic interface cancellation

Multiplying by test functions and integrating by parts, the coupled steady problem reads: find $\mathbf{u} \in V_g$, $p \in Q$, $T_f \in S_f$, $\mathbf{d} \in V_s$, $T_s \in S_s$ such that

$$\hat{a}_f(\mathbf{u},\mathbf{v}) + b(\mathbf{v},p) + c(\mathbf{u};\mathbf{u},\mathbf{v}) = \int_\Gamma \sigma_f(\mathbf{u},p)\mathbf{n}\cdot\mathbf{v}\,\mathrm{d}s + \int_{\Gamma_{\text{out}}} \sigma_f\mathbf{n}\cdot\mathbf{v}\,\mathrm{d}s + \langle\text{data}\rangle, \qquad b(\mathbf{u},q) = 0 \;\; \forall\, q, \tag{8.1}$$

$$\rho c_p\, a_T(T_f,\chi) + a_{kf}(T_f,\chi) = -\int_\Gamma k_f \nabla T_f\cdot\mathbf{n}\,\chi\,\mathrm{d}s, \tag{8.2}$$

$$\int_{\Omega_s^0} \boldsymbol{\sigma}(\mathbf{d},\theta) : \boldsymbol{\varepsilon}(\boldsymbol{\varphi})\,\mathrm{d}X = \int_\Gamma \boldsymbol{\sigma}\mathbf{n}\cdot\boldsymbol{\varphi}\,\mathrm{d}s, \tag{8.3}$$

$$a_{ks}(T_s,\chi) = \int_{\Gamma_h^0} q''\chi\,\mathrm{d}s - \int_\Gamma k_s\nabla T_s\cdot\mathbf{n}\,\chi\,\mathrm{d}s, \tag{8.4}$$

for all test functions, with $\hat{a}_f(\mathbf{u},\mathbf{v}) = 2\mu\int_{\Omega_f}\mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{v})\,\mathrm{d}x$, $b(\mathbf{v},p) = -\int_{\Omega_f} p\nabla\cdot\mathbf{v}\,\mathrm{d}x$, and $\sigma_f = -p\mathbf{I} + 2\mu\mathbf{D}(\mathbf{u})$ the fluid Cauchy stress. The interface mechanism is worth stating exactly, because it is the load-bearing sign argument of the coupling. After integration by parts, the fluid residual contains the interface term $-\int_\Gamma \sigma_f\mathbf{n}\cdot\mathbf{v}\,\mathrm{d}s$ (with $\mathbf{n}$ the outward normal of Ω_f), while the solid residual contains $+\int_\Gamma \sigma_s\mathbf{n}\cdot\boldsymbol{\varphi}\,\mathrm{d}s$ (the solid's outward normal on Γ is $-\mathbf{n}$). Under the traction equality (5.2), $\sigma_s\mathbf{n} = \sigma_f\mathbf{n}$, and with identical test functions on a conforming interface mesh the two terms enter the assembled residual with opposite signs and cancel identically; likewise the flux terms $-\int_\Gamma k_f\nabla T_f\cdot\mathbf{n}\,\chi\,\mathrm{d}s$ and $+\int_\Gamma k_s\nabla T_s\cdot\mathbf{n}\,\chi\,\mathrm{d}s$ cancel under (5.4). The monolithic statement therefore contains *no* modelled interface quantity, which removes one class of coupling error by construction. Three conditions are required for the cancellation to survive discretization, and all three are part of the specification: (i) the single-normal convention of Section 2.3; (ii) a conforming interface mesh, so that traction and temperature transfer without projection; and (iii) identical quadrature and the same isoparametric representation of the curved P2 interface on both sides, so that the two boundary integrals are evaluated on the same discrete surface — this quadrature-consistency audit is part of tests V-B and V-D. The outflow term in (8.1) is the do-nothing natural condition of Section 5.2 and does not cancel.

8.3 Solution procedure for the steady problem

The domain map $\widehat{\mathbf{w}}$ is defined as the solution of a pseudo-elastic problem on Ω_f^0 with coefficient graded to stiffen small elements near the interface, Dirichlet data $\widehat{\mathbf{w}} = \mathbf{d}$ on Γ_{if}^0 and $\widehat{\mathbf{w}} = \mathbf{0}$ on rigid boundaries – the standard solid-based mesh-motion device of deformable-domain computations [13]. The steady coupled state is obtained by the following fixed-point iteration on the geometry:

1. solve the fluid subproblem on the current Ω_f by exact Newton linearization of (8.1) (the advection derivative is linearized analytically);
2. transfer pressure-dominated traction and interface temperature requirements to the solid;
3. solve the thermoelastic subproblem (8.3)–(8.4) at the reduced (frozen-force) von Kármán level specified by Section 4.7 (linear thermoelasticity augmented by the geometric-stiffness contribution of the current membrane state; the membrane force is re-evaluated from the computed state after each solve, so the consistency update (4.11) is recovered at the fixed point; one direct solve per iteration);
4. solve the mesh-extension problem, update Ω_f, and recompute all geometry-dependent data (normals, Jacobians, quadrature);
5. under-relax the interface displacement with a documented factor ω (tuned once and reported per run);

6. iterate to a smallness of the interface displacement increment, $\|\mathbf{d}^{(k+1)}-\mathbf{d}^{(k)}\|_\infty \le 10^{-6}\, w_{\max}$ (achieved values reported per run; the earlier figure $10^{-8}H_0$ sat below achievable fixed-point accuracy and is superseded), together with residual smallness of all four subproblem statements.

The escalation trigger to the monolithic path is numeric: escalation occurs when $N_\theta^c/N_{\mathrm{cr}} > 0.5$ at any iterate, or when the geometry increment stagnates (four consecutive iterations with reduction of the increment by less than a factor of two). At the baseline ($N_\theta^c/N_{\mathrm{cr}} = 0.275$) the fixed point remains the specified path. If it nevertheless stagnates or shows loss of convergence near the buckling margin, the recorded escalation is a monolithic Newton solve including consistent shape-derivative terms; pseudo-transient continuation is the fallback. Linear systems are factorized directly (the two-dimensional problem size at the target meshes is of order 10^4–10^5 degrees of freedom); iterative solvers are a scaling option, not a baseline assumption.

8.4 Mesh convergence and quality controls

Three nested uniformly refined meshes are used per configuration; observed orders of convergence and grid-convergence indices are reported for the monitored quantities mean Nusselt number, maximum interface deflection, and pressure drop. A configuration enters the reported set only if the fine-to-medium difference is within a pre-registered tolerance (1%, tightened to 0.5% for any headline figure). In addition to the residuals, every computed run must satisfy, and tabulate, the following conservation and consistency checks, which follow directly from the formulation: (i) *global enthalpy balance*, $|\int_{\Gamma_h} q''\,\mathrm{d}s - \dot{m}' c_p (T_{b,\mathrm{out}} - T_{b,\mathrm{in}})| \le 10^{-3} Q'$, evaluated on the deformed geometry; (ii) *interface flux-jump norm*, $\|[k\partial_n T]\|_{L^2(\Gamma)}/\|q''\| \le 10^{-3}$; (iii) *load-case consistency*: mechanism attribution is performed by separate solutions (thermal only, traction only, both) rather than by adding fields; exact additivity of the linear model is verified on a detuned case with $N_\theta^c/N_{\mathrm{cr}} \le 0.05$ (Section 8.2, Chapter 9); (iv) *mean-stress identity* (3.8a,b) pointwise within interpolation accuracy; (v) *regime guards* A-1–A-6 of Chapter 6 recomputed from the computed fields.

8.5 Post-processing definitions

The bulk temperature $T_b(x)$ is mass-flux-weighted; the wall heat flux is $q_w(x) = -k_f \nabla T_f \cdot \mathbf{n}|_\Gamma$; the local heat-transfer coefficient and Nusselt number are

$$h(x) = \frac{q_w(x)}{T_w(x) - T_b(x)}, \qquad \mathrm{Nu}(x) = \frac{2H_0 h(x)}{k_f}, \tag{8.5}$$

with Nu singular where $T_w \to T_b$ near the start of the heated segment; that singular behaviour is documented and plotted rather than suppressed. The areal mean is taken over the heated length. Pressure drop is measured between area-averaged pressures on the two rigid end sections. Solid post-processing exports complete Cauchy components (never the mechanical part alone) and forms σ_m from (3.8a) with the identity (3.8b) exercised as a runtime assertion.

8.6 Computational feasibility note

All prescribed components (Taylor–Hood assembly, conforming interface mesh, direct sparse factorization, mesh-extension mapping, fixed-point loop) are standard and require no specialised CFD infrastructure; the dominant implementation risk is not algorithmic but metrological: whether the checks (i)–(v) are exercised, reported, and used as exclusions, as required in Chapter 9. The verification of the baseline formulation against an independently implemented model (an open-source finite-element platform) is part of the validation methodology (Section 9.5).

Chapter 9

Verification and validation methodology

This chapter specifies how each part of the future implementation will be checked before any coupled production computation is undertaken. Benchmarks are organised as a hierarchy from exact solutions to literature benchmarks; each entry states the exact configuration, the benchmark quantity with its normalisation, the source, and the applicability to the present solver. "Direct" means the benchmark exercises the same equations and boundary conditions as a verified component; "analogous" means it tests a mechanism under different conditions and is labelled as such.

9.1 Benchmark hierarchy

Table 9.1. Verification hierarchy V-A. . . V-F.

ID	Exact problem	Reference quantity / value	Source	Applicability and tolerance
V-A	Plane Poiseuille flow in the slit, fully developed, rigid walls	Profile $u(y) = 6U(y/H_0)(1 - y/H_0)$; $f_D\mathrm{Re}_{D_h} = 96$ (Darcy)	Analytic; tabulated in [2]	Direct. L^2 velocity error to machine precision on the fine mesh; friction within 10^{-4} relative
V-B1	Rigid slit, both walls at (i) uniform T_w, (ii) equal uniform flux	$\mathrm{Nu}_T = 7.541$; $\mathrm{Nu}_{H2} = 8.235$ (Nu on $T_w - T_b$, basis D_h)	[2]	Direct thermal family, two-sided heating only
V-B2	Rigid slit, one wall uniform flux / adiabatic other (H1); one wall uniform T (T1)	$\mathrm{Nu}_{H1} = 5.385$; $\mathrm{Nu}_{T1} \approx 4.861$	[2]; T1 digits corroborated by independent table reproductions, primary-table confirmation still the practice standard (App. B, C-1)	Direct: this is the geometry's own rigid family
V-C1a	Fully constrained body, uniform rise θ	$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = -\beta\theta$; $\sigma_m = -3K\alpha\theta$	Derived here, (4.2)	Direct, to quadrature precision

ID	Problem	Value	Source	Applicability/tolerance
V-C1b	Axially restrained plane-strain strip, free faces, uniform θ	$\sigma_{xx} = \sigma_{zz} = -E\alpha\theta/(1-\nu)$; $\sigma_m = -2E\alpha\theta/[3(1-\nu)]$	Derived here, (4.3) (corrected pairing, App. B, D-12)	Direct
V-C2	Free plane-strain strip: (i) uniform θ; (ii) linear through-thickness gradient	(i) $\sigma_m = -E\alpha\theta/3$, (4.4); (ii) $\kappa = (1+\nu)\alpha\Delta T_t/d$, (4.5)	Derived here; classical theory [8]	Direct; curvature to 0.1%
V-C3	Pinned–pinned plane-strain strip with constant eigencurvature, slender specimen $L/d = 25$: (i) zero membrane force; (ii) subcritical compressive force	(i) $w_{\max} = \kappa_{\text{th}} L^2/8$, (4.6); (ii) (4.10), amplification 1.39 at $\mu L = 1.65$	Derived here; beam theory [16]	Direct; 1% (documented shear estimate)
V-D	The frozen geometry in the rigid-wall limit (deformation suppressed by scaling E), single-side conjugate heating	Developing $\mathrm{Nu}(x)$ versus published conjugate thick-wall channel data at matched R_k, D_r, Pe	The flattened thick-wall channel series of [18] (steady laminar water, single-side heating, thickness and conductivity ratios) is the designated matching target; exact matching points to be extracted (App. B, C-2)	Direct after matching; analogous otherwise. Also in-house enthalpy identity
V-E	Steady bulge of an elastic-wall shallow slit channel under imposed Δp, small deflection, linear elastic wall	Mid-deflection and Q–Δp relation versus lubrication/plate-theory predictions	Theory class: [15]; the shallow-channel assumption holds here ($H_0/L = 1/60$); the benchmark data set is to be constructed for the present geometry (App. B, C-3)	Direct for the fluid–elastic coupling of the steady code
V-F	Per-run quality controls of Section 8.4	residual- and balance-based pass/fail	Specification of this document	Mandatory for every reported configuration
V-G	One case computed on the mapped domain *and* on an independently generated body-fitted mesh of the same deformed geometry	Field-level difference (deflection, $\overline{\mathrm{Nu}}$, Δp) isolating the mesh-map contribution	Specification of this document	Mandatory once for the coupled solver, at the anchor operating point

Because the baseline Péclet number is Pe = 348, every thermal case above is run with the stabilized transport discretization of Chapter 8. The acceptance criterion in each case is that the stabilized fine-mesh value reproduce the benchmark within the stated tolerance with no under- or overshoot of the temperature field beyond 0.1% of the applied range; in addition, the stabilization contribution is reported as the stabilized-minus-Galerkin difference on the finest mesh on which the unstabilized solve remains bounded, so that the size of the numerical, as opposed to physical, contribution is on record. Energy-stabilization requirements beyond SUPG are not invoked at the baseline parameters (cell Péclet well above unity is handled by the upwinding itself); any re-arming of stronger stabilization would be a documented change of the discretization specification.

9.2 Benchmarks explicitly not used

Turek and Hron's widely used FSI benchmark [9] consists of *large-displacement* configurations (the steady FSI1 set and the transient FSI2/FSI3 sets) with hyperelastic flag behaviour; it does not exercise, and cannot validate, a steady small-strain linear formulation, and agreement with it would not be evidence for or against the present solver. If a transient ALE extension (Section 11.1) is later undertaken, that benchmark becomes an appropriate acceptance test and is reserved for it; such an extension must also state its relation to the steady formulation explicitly — initial condition taken as the converged steady solution, or start-up transient reported to stationarity — rather than inherit the steady results silently (register D-10). Likewise, the two-sided-heating Nusselt values of V-B1 must not be quoted as targets for the single-heated-wall geometry: one-sided ranges are governed by the H1/T1 family [2].

9.3 Cross-component verification items

Two items cut across the hierarchy. Item **B-1** (buoyancy, cf. Section 7.3): at the baseline the exclusion is now closed by the corrected order-of-magnitude estimate ($\mathrm{Ri} = 1.7\times10^{-3}$, three orders below unity); the item is retained as a regime guard, re-armed only if a parameter-study point drives Ri above 10^{-2}, in which case a Boussinesq-enabled case at that point is required with the neutral-density difference reported. Item **C-4** (second-solver replication): an independent implementation of one anchor operating point in an open-source finite-element platform (FEniCS is the default, requiring no licence), with field-level differences reported.

9.4 Staged implementation decisions

The verifications are decoupled from the coupling ambition in a staged manner:

Component	Conditions recorded for proceeding	Current status
Rigid-fluid solver (V-A)	none	**may proceed**
Rigid conjugate thermal solver (V-B)	item C-1 closed before quoting T1 values	**may proceed**
Thermoelastic solver (V-C)	unit-test expectations of V-C1a/b, V-C2 as corrected here (D-2, D-12) and the beam–column target of V-C3	**may proceed, conditional**
Coupled production computations	all of: V-D evidence obtained (C-2); V-E data set constructed (C-3); buoyancy check B-1 executed and accepted; property freeze C-6; regime guards implemented as runtime assertions	**deferred (HOLD)**
Positioning statements in any manuscript	full-document novelty search completed (C-7)	**deferred (HOLD)**

The composite gates that release the HOLD state, with their exact acceptance conditions, are preserved verbatim in the audit archive (`audit/HFF_pre_coding_checklist.md`) and summarised in Appendix B. Proceedings under "may proceed" are solver-verification exercises only and do not authorise coupled computations.

9.5 Independent replication

The cross-component item C-4 above is restated here as methodology rather than gate: a second, independently implemented model of one anchor operating point, comparing interface deflection, mean Nusselt number, and pressure drop field-wise rather than by verbal agreement. Discrepancies are reported quantitatively and resolved to identifiable causes before production runs begin.

Chapter 10

Literature review, research gap, and proposed contribution

10.1 Lines of research neighbouring the present model

Flow-load-driven flexible-wall FSI with heat transfer. Soti, Bhardwaj, and Sheridan [4] computed two-way coupled interaction of laminar channel flow with a flexible thin plate including advection–diffusion heat transfer, showing that flow-induced deformation passively raises heat transfer; the deformation is traction-driven and the structures thermally passive. The cavity studies of Alsabery et al. [14] and Saleh et al. [5] extended transient FSI with natural convection in heated cavities with flexible walls, again with deformation driven by fluid loading.

Thermo-fluid–structure interaction with heated flexible structures. The recent work of Kumar, Assam, and Prabhakaran [7] computes nonlinear thermo-FSI of two interacting cantilever plates in a differentially heated cavity – the closest relative of the present problem in its closure of the thermal loop, though the configuration is an unconfined cavity with beam-like structures.

Steady compliant-channel theory. Christov and co-workers [15] established the relation between flow rate and pressure drop in deformable shallow microfluidic channels, providing the analytical class used for the steady elastic-coupling benchmark V-E. The channel there is deformed by fluid pressure alone; thermal bowing does not appear.

Adjacent physics. The stability of stably stratified horizontal channel flow, including deteriorated heat transfer under stable stratification, has been characterised by direct numerical simulation in [12]; that work marks the opposite limit of the buoyancy question — the baseline Richardson number of the present problem lies three orders of magnitude below the stratification-relevant range (Chapter 7) — and is not itself a conjugate-coupling study. The thermoelastic theory underlying wall curvature under imposed gradients is classical [8, 17], as are the buckling limits [16] and the rigid-duct thermal entry results [2].

10.2 Research gap and its stated bounds

The search underlying this document (Appendix B, item C-7, with its documented limitations) did not identify a study that combines the following elements in one computed model: (i) *thermal eigen-curvature of the wall itself* – the through-thickness gradient of a heated channel wall – as the principal designed deformation mechanism, rather than traction-driven flexibility; (ii) a *closed two-way steady loop* in which the deformed passage modifies the conjugate heat transfer whose boundary condition produces the deformation; (iii) an analysis organised in the mechanism parameters (6.6), Γ_p, and M_b, with explicit hydraulic-driving-condition contrast (fixed flow rate versus fixed pressure drop); and (iv) verification structured along analytical limits, rigid-wall conjugate benchmarks, and a steady elastic-coupling benchmark of the Christov class, rather than transient FSI suites.

This is a *bounded* positioning, not a claim of priority: thermo-FSI as a class exists [7]; flexible-wall non-isothermal FSI exists [4, 5, 14]; and compliance benchmarks of deformable channels exist [15]. What the present configuration changes is the mechanism (thermal bowing in the wall), the flow class (confined forced-convection slit at $\mathrm{Re}_{D_h} \leq 100$), and the steady formulation class. A systematic database pass (Scopus/Web of Science) remains open (Appendix B, C-7), and no wording stronger than the bounded statement above is supported until it is completed.

10.3 Intended contribution of the proposed work

Within those bounds, the proposed study will contribute:

1. a verified, monolithic-compatible formulation of a thermally self-deforming conjugate channel with exact interface cancellation (Section 8.2);
2. an analytical regime description (6.6) together with design-relevant ceilings (wall-stress cap of bound A-6, buckling (4.8), deformation ratio (4.16)), which orient any future parameter study before a single simulation is run;
3. a computed map of the deformation–heat-transfer feedback in the (Σ_κ, R_k, D_r, driving-condition) space, including mechanism attribution between thermal and traction contributions by the load-case decomposition of (5.5) – all to be obtained only after the verification hierarchy of Chapter 9 is satisfied.

The third item is prospective and is stated as future work: it cannot be reported, characterised, or even conjectured quantitatively before the computations exist.

Chapter 11

Limitations and concluding remarks on the formulation

11.1 Model limitations

The following limitations are inherent to the formulation as posed and accompany any future use of the model.

(1) Physical modelling limits. The wall is treated as homogeneous, isotropic, linearly thermoelastic (A3); unfilled PEEK is mildly anisotropic and its glass-transition-modified behaviour at temperatures approaching or exceeding its operating limits is not captured. Creep and stress relaxation are excluded, so the formulation describes the short-time-scale steady state and not hours-long drift. Contact resistance at the interface is neglected (A8). Constant fluid properties are self-consistent at the baseline but set an upper envelope on thermal load through bound A-4.

(2) Buoyancy. As discussed in Section 7.3, the corrected length-consistent estimate gives $\mathrm{Ri} = 1.7 \times 10^{-3}$ at the baseline, and the constant-density closure stands on that estimate. The disposition carries its own guard: the question re-opens at any parameter-study point with $\mathrm{Ri} > 10^{-2}$, the Boussinesq term remains available in the weak form, and the estimate itself is recomputed from computed fields in every reported run (regime guard A-1–A-6 audit).

(3) Support and geometric ideality. The pinned supports of A9 are idealizations; any real compliance at the supports changes both the thermal bowing and the buckling ceiling, and the clamped comparison is recorded in the audit register. The full-edge pin idealization itself carries an $O(d/L)$ stiffening relative to ideal point pins — a few per cent at the baseline aspect ratio — closed by the one-time numerical check of Chapter 9. Entry-length estimates for the hydrodynamic and thermal development assume the engineering constants used in Table 7.1 and are replaced by computed developing solutions in the future.

(4) Reduced wall equation. Equation (3.9) and the beam–column extension (4.9) are design relations at the reduced (frozen-force) von Kármán level. The reduction carries a quantified, one-sided error: at the baseline it overpredicts the self-consistent deflection by 2.3% (0.438 against $0.428\,H_0$), the error stays below $\approx 4\%$ inside the admissible envelope ($N_\theta^c/N_{\mathrm{cr}} \leq 0.33$), and it grows rapidly beyond $N_\theta^c/N_{\mathrm{cr}} \approx 0.5$, where the escalation criterion of Chapter 8 requires the consistent force update (4.11); the buckling margin is unaffected. None of the coupled-model conclusions to be obtained computationally depend on this accuracy beyond its stated order, and the wall model itself is a verification target (test V-C3).

(5) Literature coverage. The positioning of Chapter 10 is conditional on completion of the systematic search (Appendix B, C-7); several bibliographic records used here required manual extraction and one, [6], remains bibliographically incomplete (Appendix B).

(6) Data freeze. The numerical values of Table 7.1 that depend on the solid properties or on β_f assume datasheet-representative numbers pending a fixed datasheet (C-6); only the dimensionless structure, not the point values, is data-independent.

11.2 Concluding remarks

The formulation presented here is complete as a mathematical statement: field equations, constitutive relations, and boundary data are closed; the plane-strain reduction and the associated design relations are derived rather than asserted; the dimensionless structure of the regime is identified from the equations; and a baseline operating point is established arithmetically. The principal open questions are not mathematical but evidentiary: the bibliography items listed in Appendix B, the property freeze, and the construction of the two missing benchmark data sets; the buoyancy question is closed at the baseline by estimate and guarded by a regime check (Chapter 7). The coupled computations themselves are deliberately deferred until these items close; the isolated solver verifications may proceed under their stated conditions. In that sense the present document is a research formulation in the strict meaning of the word: everything stated is either established theory with a source, a derivation reproduced in full, or an explicit agenda item with named conditions.

Appendix A

Notational conventions and arithmetic calculations

A.1 Notation and units

Symbol	Meaning	Unit
x, y	current coordinates in Ω_f	m
X, Y	reference coordinates	m
$\mathbf{u} = (u, v)$	fluid velocity	$\mathrm{m\,s^{-1}}$
U	mean inlet velocity	$\mathrm{m\,s^{-1}}$
p	fluid pressure	Pa
T_f, T_s	fluid, solid temperatures	K
$\theta = T_s - T_{\mathrm{ref}}$	solid temperature rise	K
$\mathbf{d} = (d_1, d_2)$	solid displacement	m
$w(x) = -d_2\vert_{Y=H_0}$	interface deflection, positive into the channel	m
$\widehat{\mathbf{w}}$	mesh-displacement extension into Ω_f^0	m
$\boldsymbol{\sigma}, \boldsymbol{\varepsilon}$	Cauchy stress, infinitesimal strain	Pa, –
$\sigma_m = \mathrm{tr}\,\boldsymbol{\sigma}/3$	mean Cauchy stress	Pa
$\mathbf{D}$	rate-of-deformation tensor, fluid	$\mathrm{s^{-1}}$
E, ν, α	Young's modulus, Poisson ratio, thermal expansion coefficient	Pa, –, $\mathrm{K^{-1}}$
G, K	shear and bulk moduli	Pa
λ, μ_L	Lamé parameters ($\mu_L = G$)	Pa
$\beta = 3K\alpha$	thermoelastic pressure coefficient	$\mathrm{Pa\,K^{-1}}$
ρ, c_p, k_f, β_f	fluid density, specific heat, conductivity, expansion coefficient	SI
ρ_s, c_s, k_s	solid density, specific heat, conductivity	SI
$\mathbf{n}$	interface unit normal, from fluid into solid	–
q''	imposed heat flux on Γ_h^0	$\mathrm{W\,m^{-2}}$
$T_{\mathrm{ref}}, T_{\mathrm{in}}$	reference and inlet temperatures	K
H_0, D_h, d, L, L_e, L_o	heights, hydraulic diameter, thickness, lengths	m
τ_w	wall shear stress	Pa
T_b	mass-flux-weighted bulk temperature	K
$\dot{m}'$	mass flow per unit depth, $\rho U H_0$	$\mathrm{kg\,s^{-1}\,m^{-1}}$
f_D	Darcy friction factor	–
Nu	hD_h/k_f with h on $T_w - T_b$	–
ΔT_t	through-wall temperature drop, $q'' d/k_s$	K
κ_{th}	free thermal eigen-curvature	$\mathrm{m^{-1}}$
D_s	flexural rigidity per unit depth	N m
$N_\theta, N_{\mathrm{cr}}$	thermal membrane force, pinned critical load	N/m

Symbol	Meaning	Unit
$\bar{p}$	datum-consistent mean gauge pressure on the strip	Pa
μ	dynamic viscosity (fluid); also beam–column parameter $\mu^2 = N_\theta^c/D_s$	Pa s; m^{-1}
Γ_p	thermal-to-pressure deflection ratio	–
ξ	thermoelastic coupling number	–

A.2 Arithmetic cells (independent recomputation)

Each cell gives expression, substituted inputs, and result; cells are numbered for citation from the audit material.

A1. Pressure drop. $\Delta p = 12(10^{-3})(5\times10^{-2})(3\times10^{-2})/(2.5\times10^{-7}) = 72$ Pa. Independent check: with $f_D = 96/50 = 1.92$, $f_D(L/D_h)\rho U^2/2 = 1.92 \times 30 \times 1.25 = 72$ Pa; force balance $\Delta p H_0 = 3.6 \times 10^{-2}$ Pa m $= 2\tau_w L$ with $\tau_w = 6\mu U/H_0 = 0.6$ Pa.

A2. Mean-stress identity, restrained strip. With $e = \alpha\theta(1+\nu)/(1-\nu)$, $K(e-3\alpha\theta) = K\alpha\theta(4\nu-2)/(1-\nu) = 2E\alpha\theta(2\nu-1)/[3(1-2\nu)(1-\nu)] = -2E\alpha\theta/[3(1-\nu)]$, equal to the stress form of (4.3).

A3. Thermal curvature and bow (zero membrane force). $\kappa_{\mathrm{th}} = 1.4(5\times10^{-5})(5\times10^{3})/0.25 = 1.4$ m^{-1}; $w_{\max} = 1.4(9\times10^{-4})/8 = 1.575\times10^{-4}$ m. With the baseline membrane force the frozen-force beam–column solution (cell A9) gives $w_{\max} = 2.19\times10^{-4}$ m (self-consistent 2.14×10^{-4} m); bending fibre strain $(\kappa_{\mathrm{geo}} - \kappa_{\mathrm{th}})d/2 = 9.9\times10^{-4}$; arc-stretching term $\frac{1}{2}\langle w'^2\rangle = 1.8\times10^{-4}$ (circular-arc estimate $\kappa_{\mathrm{geo}}^2 L^2/24 = 1.6\times10^{-4}$).

A4. Buckling catalog. With denominator $12\alpha(1+\nu)L^2 = 7.56\times10^{-7}$, $d = \{1, 1.5, 2, 2.5, 3\}$ mm gives $\bar{T}_{\mathrm{cr}} = \{13, 29, 52, 82, 117\}$ K at $L = 30$ mm (rounded).

A5. Membrane force and margin. N_θ at $\bar{\theta} = 32.3$ K: -2.91×10^4 N/m; $N_{\mathrm{cr}}(d = 3$ mm$) = 9.87\times9.64/9\times10^{-4} = 1.057\times10^5$ N/m; $M_b = 117.5/32.3 = 3.64$.

A6. Coupling number. $\xi = (9\times10^5)^2(293)/[(7.7\times10^9)(1.43\times10^6)] = 2.2\times10^{-2}$.

A7. Time scales. $t_{\mathrm{wave}} = 3\times10^{-3}\sqrt{1.3\times10^3/3.6\times10^9} = 1.8\times10^{-6}$ s; $\kappa_s = k_s/(\rho_s c_s) = 1.75\times10^{-7}$ m^2 s^{-1}, $t_s = 9\times10^{-6}/1.75\times10^{-7} = 51$ s; $t_f = 2.5\times10^{-7}/1.44\times10^{-7} = 1.7$ s.

A8. Deformation ratio (datum-consistent load). With $\bar{p} = \frac{1}{2}[p(0)+p(L)] = 84$ Pa (pressure datum of Section 4.8), $w_p = 5(84)(3\times10^{-2})^4/[384(9.6)] = 9.2\times10^{-8}$ m; $\Gamma_p = 2.19\times10^{-4}/9.2\times10^{-8} = 2.4\times10^3$.

A9. Beam–column bowing at the baseline (frozen force). $\mu^2 = N_\theta^c/D_s = 2.9\times10^4/9.6 = 3.0\times10^3$ m^{-2}, $\mu = 55$ m^{-1}, $\mu L = 1.65$; exact solution (4.10) gives $w_{\max} = 2.19\times10^{-4}$ m $(= 0.438H_0)$, amplification 1.39 over the parabolic bow, against the leading-order factor $1/(1-N_\theta^c/N_{\mathrm{cr}}) = 1/0.725 = 1.38$. Mid-span moment $M_{\mathrm{mid}} = -6.4$ N m per unit depth; $\sigma_{\mathrm{bend}} = 6|M|/d^2 = 4.2$ MPa; $w'_{\max} = 0.027$ rad; blocked mechanical strain $e_0 = (1+\nu)\alpha\bar{\theta} = 2.26\times10^{-3}$; arc-stretching term $\frac{1}{2}\langle w'^2\rangle = 1.8\times10^{-4}$ (circular-arc estimate 1.6×10^{-4}): first-iterate force relaxation 6.1%, self-consistent 5.9% (4.11); total mechanical surface strain $\approx 5.4\times10^{-3}$. Self-consistent comparison values: $w_{\max} = 2.14\times10^{-4}$ m $(0.428H_0$; frozen force overpredicts by 2.3%, conservative), $M_{\mathrm{mid}} = -5.9$ N m, $\sigma_{\mathrm{bend}} = 3.9$ MPa, $\sigma_{\max} = 13.6$ MPa $= 0.151\sigma_y$; buckling margin $M_b = 3.64$ exact (relaxation vanishes at onset). Ceiling re-solve on the frozen-force relation: A-3 mesh bound at $N_\theta^c/N_{\mathrm{cr}} = 0.302$ ($q'' \approx 5.5\times10^3$ W m^{-2}), A-6 stress cap at 0.327 ($\approx 5.9\times10^3$), $M_b = 2$ at 0.5 ($\approx 9.1\times10^3$).

A.3 Dimensionless dictionary

Group	Definition	Baseline value
Re_{D_h}	$2\rho U H_0/\mu$ (Darcy basis, $D_h = 2H_0$)	50
Re_{H_0}	$\rho U H_0/\mu$	25
Pr; Pe	$\mu c_p/k_f$; $\mathrm{Re}_{D_h}\mathrm{Pr}$	6.97; 348
Gr; Ri	$g\beta_f \Delta T^* H_0^3/\nu_{\mathrm{kin}}^2$; $\mathrm{Gr}/\mathrm{Re}_{H_0}^2$, (6.3)	1.07; 1.7×10^{-3} (3.4×10^{-3} on the D_h basis)
Br	$\mu U^2/(k_f \Delta T^*)$	1.0×10^{-6}
R_k; D_r; Λ	k_s/k_f; d/H_0; L/H_0	0.42; 6; 60
Σ_κ	$\alpha q'' H_0/k_s$	5×10^{-4}
$w_{\max}^{\mathrm{lin}}/H_0$	$(1+\nu)\Sigma_\kappa \Lambda^2/8$, (6.6)	0.315
$w_{\max}/H_0$	beam–column, (4.12)	0.438
Γ_p	$48\kappa_{\mathrm{th}} D_s/(5\bar{p}L^2)$, $\bar{p}$ from (4.15)	2.4×10^3
M_b	$\bar{T}_{\mathrm{cr}}/\bar{\theta}$	3.64
ξ	$\beta^2 T_0/[(\lambda + 2\mu_L)\rho_s c_s]$	2.2×10^{-2}

Appendix B

Audit record: discrepancies, evidence status, and open checks

This appendix is deliberately separated from the research exposition of Chapters 1–11. It records, for independent review, the discrepancies found during preparation of the formulation, the consolidated status of evidence, the bibliographic and data items still open, and the conditions attached to the staged implementation decisions. The six source documents from which this record derives are preserved verbatim in the `audit/` folder of this package.

B.1 Discrepancy register

Where working drafts disagreed with each other or with the derivations presented here, neither value was silently preferred; the arithmetic was redone and both positions are recorded.

Table B.1. Discrepancy register D-1 … D-20 (D-14–D-19 added at the second reconciliation pass; D-20 at the Stage-3 synchronization).

ID	Divergence	Resolution adopted (with basis)	Status
D-1	Plane-strain constitutive forms across drafts differed in the treatment of σ_{zz} (occasionally set to zero)	Rigorous $\varepsilon_{zz} = 0$ elimination, (3.6), (4.1)	resolved
D-2	Free uniform-expansion mean-stress expectation quoted as $\sigma_m = 0$	Plane-strain value $\sigma_m = -E\alpha\theta/3$, (4.4)	resolved
D-3	Early conceptual framing spoke of pressure-augmented compliance and buckling-linked deformation as the mechanism	Pure thermal-eigen-curvature mechanism adopted; supports and buckling treated as regime constraints only	resolved
D-4	Critical temperature derived with the clamped–clamped eigenvalue (≈ 200 K) versus the pinned design	Pinned value (4.8); catalog in cell A4	resolved, design-locked

ID	Divergence	Resolution	Status
D-5	Pressure-drop expression printed with H_0^3 (dimensionally Pa/m), giving $\Delta p \sim 1.4 \times 10^5$ Pa and $\Gamma_p \simeq 3$	Correct (4.13); $\Delta p = 72$ Pa at baseline (cell A1); Γ_p now governed by D-13	resolved
D-6	Friction-factor basis left implicit between text and tables	Darcy $f_D \mathrm{Re}_{D_h} = 96$; basis must be printed on computed outputs	resolved as convention
D-7	Mean wall rise $\bar{\theta}$ treated variously as exact linear-drop value and as order-one estimate	Estimate with component-wise accuracy (Table 7.1); sweeps to report ranges	resolved as convention
D-8	Buoyancy exclusion first justified by $\mathrm{Ri} = 0.43 < 1$ alone (value itself defective: mixed length basis and mistranscribed powers)	Length-consistent re-evaluation: $\mathrm{Ri} = \mathrm{Gr}(H_0)/\mathrm{Re}(H_0)^2 = 1.7 \times 10^{-3}$ (3.4×10^{-3} on the D_h basis); exclusion by three-order magnitude margin; regime guard $\mathrm{Ri} > 10^{-2}$ re-arms a Boussinesq-enabled case; dormant Boussinesq term kept in the weak form	resolved (second pass)
D-9	Constant fluid properties asserted project-wide	Self-consistency shown at baseline only (fluid-side excursions 1–3 K); bound A-4 enforced	resolved at baseline, bounded
D-10	Positioning note "no two-way T-FSI exists"	T-FSI exists as a class [5, 7]; scoped claim of Ch. 10 replaces it	resolved
D-11	Admissibility bound $\lvert w\rvert/H_0 \leq 0.2$ conflicted with the baseline value 0.315; membrane strain figures of different origin mixed	Strain-based bounds (A-3, re-derived at the beam–column level, D-15): bending fibre 9.9×10^{-4}, total mechanical surface 5.4×10^{-3}, arc-stretching 1.8×10^{-4} (circular-arc estimate 1.6×10^{-4}), slope 0.027 rad; mesh-generation bound 0.5 with positive Jacobians	resolved
D-12	Restrained-strip benchmark paired σ_{xx} of one restraint case with σ_m of another	Cases separated: (4.2) and (4.3)	resolved
D-13	Pinned–pinned uniform-load deflection written with the clamped–clamped coefficient 1/384	Correct pinned coefficient 5/384, (4.14); load level re-examined in D-16; conclusion (thermal dominance) unchanged	resolved

ID	Divergence	Resolution	Status
D-14	Gr quoted as 1.1×10^3, Ri as 0.43, Br as 1.0×10^{-3} (three powers-of-ten defects in one cluster; an intermediate audit corrected Ri to 4.3×10^{-4} on the H_0 basis but retained the mixed-division definition)	All three recomputed with full arithmetic (cells A6a, dictionary): Gr = 1.07; Ri = $\mathrm{Gr}(H_0)/\mathrm{Re}(H_0)^2 = 1.7 \times 10^{-3}$ on the single-length basis, with the basis stated on the group; Br = 1.0×10^{-6}	resolved (second pass)
D-15	Baseline bowing quoted from the zero-force parabola although the strip's own thermal membrane force is finite ($N_\theta^c/N_{\mathrm{cr}} = 0.275$); binding mechanical constraint stated as buckling	Beam–column relation (4.10) adopted: $w_{\max} = 2.19\times10^{-4}$ m $= 0.438H_0$ (amplification 1.39; cell A9); A-3 strain bounds rederived at that level; constraint ordering corrected. Ceiling resolve at Stage 3 on the frozen-force relation (the second-pass interpolation used an inconsistent amplification scaling): A-3 mesh bound ($\lvert w\rvert/H_0 \le 0.5$) binds at $N_\theta^c/N_{\mathrm{cr}} = 0.302$ ($q'' \approx 5.5 \times 10^3$ W m^{-2}), the wall-stress cap at 0.327 ($\approx 5.9\times10^3$), buckling remains the outer ceiling ($M_b = 2$ at 0.5, $\approx 9.1 \times 10^3$); ordering unchanged	resolved (second pass; ceilings exact at Stage 3, D-20)
D-16	Mean hydraulic load on the strip taken as $\bar{p} = \Delta p/2 =$ 36 Pa, although the pressure *level* is fixed by the outlet datum B2 and the casing environment, not by Δp	Datum statement added to A9/B2: gauge 48–120 Pa along the strip, mean $\bar{p} = 84$ Pa (4.15); $w_p = 9.2\times10^{-8}$ m, $\Gamma_p = 2.4 \times 10^3$ (cell A8); conclusion (thermal dominance, $\Gamma_p \gg 5$) unchanged under any plausible casing datum	resolved (second pass)
D-17	Bending-stress scale quoted from the fully restrained-curvature expression $E\kappa_{\mathrm{th}}(d/2)/(1-\nu^2) = 9.0$ MPa, which corresponds to no admissible support state of the pinned strip	Beam–column mid-span moment used instead: $M_{\mathrm{mid}} = -6.4$ N m, $\sigma_{\mathrm{bend}} = 4.2$ MPa; combined stress ≈ 14 MPa $= 0.15\sigma_y$ (cell A9)	resolved (second pass)
D-18	Symbol collisions: conductivity ratio K and thickness ratio D against bulk modulus and rate-of-deformation tensor; conduction matrix K against capacity matrix in the solver equations	Renamed: R_k, D_r, K_T, C_{lump} (Ch. 6, 8)	resolved (second pass)

ID	Divergence	Resolution	Status
D-19	Admissibility bound A-6 worded "within 20% of the yield stress", which reads as $0.8\sigma_y$	Recast as design cap $\sigma_{\max} \leq 0.2\sigma_y$ with margin statement (Ch. 6)	resolved (second pass)
D-20	The beam–column equation was described as the "consistent von Kármán wall equation" and the frozen-force treatment as "self-consistent"; the second-pass ceiling estimates used an inconsistent amplification scaling; three Chapter-11 limitation updates (buoyancy, supports, reduced wall equation) were lost to a failed editing pass and absent from Revision 2	Targeted audit (beam–column consistency) reclassified the model: reduced (frozen-force) von Kármán reduction, first iterate of the compatibility update (4.11); justified with explicit limitation — over-predicts the self-consistent deflection by 2.3% at baseline (0.438 vs 0.428 H_0), error $< 4\%$ inside the admissible envelope, one-sided (conservative), buckling margin $M_b = 3.64$ exact, error grows beyond $N_\theta^c/N_{\text{cr}} \approx 0.5$ where escalation triggers; force relaxation 6.1% (first iterate), 5.9% (self-consistent); ceilings re-solved exactly (D-15); sign convention (signed resultant $N_\theta < 0$ in compression vs positive magnitude N_θ^c) stated; Chapter-11 limitations restored	resolved (Stage 3 synchronization)

B.2 Consolidated evidence status

The external claim--evidence register (`audit/HFF_claim_evidence_register.csv`, 41 entries) is summarised here with status labels restricted to this appendix:

Status in the register	Count
verified	18
verified as specification/convention	4
verified with conditions	3
derived (with assumed property inputs as noted)	2
bounded claims, recommendations, decisions	5
unresolved / data not extracted / not applicable	5
superseded (incorrect) (superseded by this document)	4
Total	41

All statements superseded by draft errors have been absorbed into Table B.1; none remains active in the equations of the present formulation. Nothing marked **unresolved** may be quoted as fact in any manuscript derived from this document.

B.3 Open checks (with named action items)

- **C-1** Digit-level extraction of the T1 Nusselt value from the primary table of [2]. Only H1 and T/H2 values are anchored at source level.
- **C-2** Extraction and geometry matching (R_k, D_r, Pe) of a published conjugate thick-wall channel Nusselt series for V-D. The flattened thick-wall channel series of [18] (steady laminar water, constant outer wall temperature, Re 50–1250, single-side heating, reported thickness- and conductivity-ratio parameters) is the designated matching target identified at the second pass; the digit-level matching points remain to be extracted.
- **C-3** Construction of the steady compliant-channel benchmark data set for V-E from the lubrication/plate-theory predictions of [15] (shallow-channel limit).
- **C-6** Freeze of the PEEK datasheet set ($E, \nu, \alpha, k_s, \rho_s, c_s$, yield band) and of β_f at fixed T_{ref}; until then, baseline numbers carry the marked dependency.
- **C-7** Systematic originality search (Scopus/WoS and full-document pass), after which the positioning of Chapter 10 may be finalised.
- **C-9** Outflow-condition sensitivity (exit length, no-backflow, datum convention) for condition B2.
- **C-12** Commitment to the second-solver platform (FEniCS default) required by Section 9.5.

B.4 Bibliographic integrity statement

Every entry in `references.bib` was verified at assembly against Crossref metadata or a publisher record; DOI, volume, pages, and authorship were confirmed at source level for all journal articles and for the book chapters cited. Two records are intentionally incomplete: [6] (title and DOI not retrievable at assembly; indexing facts verified from the publication record) and the T1 digit of [2] pending C-1. No entry contains invented fields; fields not verified are either absent or marked in `note`.

B.5 Staged decisions carried from the audit

Isolated solver verifications may proceed as specified in Chapter 9 (V-A without condition; V-B after C-1 for T1 quotations; V-C with the corrected unit-test expectations of register D-2 and D-12). Coupled production computations remain on HOLD pending C-2, C-3, C-6, and implementation of regime guards as runtime assertions (the buoyancy check B-1 is closed by the corrected estimate, D-8, and survives only as the regime guard); manuscript positioning statements remain on HOLD pending C-7. These decisions are recorded here verbatim in effect: *no* part of this document authorises coupled production runs.

B.6 Second reconciliation pass: scope and disposition

An independent second reconciliation pass re-derived every dimensionless group and design quantity of the formulation from its defining expression with explicit units and substitutions, re-derived the pinned beam–column relation from the von Kármán wall equations, and re-examined each item of the register above against both the original working values and the first-pass audit values; neither source was accepted automatically. The outcomes are recorded in rows D-14–D-19 and in the reconciliation log accompanying this document. Items resolved at the second pass: the Grashof/Richardson/Brinkman cluster (D-14), the finite-force bowing relation and the resulting constraint ordering (D-15), the hydraulic load datum (D-16), the bending-stress scale (D-17), the symbol collisions (D-18), and the A-6 wording (D-19). Items deliberately left open, with

named actions: C-1 (T1 Nusselt digits), C-2 (digit-level V-D matching), C-3 (V-E data set), C-6 (property freeze), C-7 (systematic originality search), C-9 (outflow-condition sensitivity), and C-12 (second-solver platform commitment). No discrepancy was resolved by adjusting geometry, material data, or operating conditions.

B.7 Stage-3 synchronization: beam–column model classification

A targeted consistency audit of the beam–column model (record `HFF_Reconciliation_2/HFF_beamcolumn_model_consistency.md`) confirmed the adopted equation as the *reduced (frozen-force) von Kármán* beam–column relation — the transverse member of the von Kármán system with the membrane force frozen at its blocked-state value — and quantified its error against the fully consistent (force-updated) solution: 2.3% overprediction of $w_{\max}$ at the baseline, one-sided (conservative), below 4% inside the admissible envelope, with an exact buckling margin; the error grows beyond $N_\theta^c/N_{\mathrm{cr}} \approx 0.5$, where the escalation criterion of Chapter 8 applies. Verdict: *existing model justified with an explicit limitation*; no model change. The same pass corrected the ceiling estimates (D-15), the arc-strain bookkeeping (D-11), restored three Chapter-11 limitation updates lost to a failed editing pass (D-20), and stated the axial-force sign convention (signed resultant $N_\theta < 0$ in compression; positive magnitude N_θ^c in all force ratios).

Bibliography

[1] P. Hood and C. Taylor. Navier–stokes equations using mixed interpolation. In *Finite Element Methods in Flow Problems*. University of Alabama in Huntsville Press, Huntsville, Alabama, 1974. Origin of the P2–P1 pair used here; page range not machine-verified at assembly.

[2] R. K. Shah and A. L. London. *Laminar Flow Forced Convection in Ducts: A Source Book for Compact Heat Exchanger Analytical Data*. Supplement 1 to Advances in Heat Transfer. Academic Press, New York, 1978. fRe(Darcy)=96, Nu(T)=7.541, Nu(H2)=8.235, Nu(H1)=5.385 verified; Nu(T1) digit awaits primary-table extraction (item C-1).

[3] R. Byron Bird, Warren E. Stewart, and Edwin N. Lightfoot. *Transport Phenomena*. John Wiley and Sons, New York, 1960. Source for the incompressible Navier–Stokes and energy equations in the form used here.

[4] Atul Kumar Soti, Rajneesh Bhardwaj, and John Sheridan. Flow-induced deformation of a flexible thin structure as manifestation of heat transfer enhancement. *International Journal of Heat and Mass Transfer*, 84:1070–1081, 2015. doi: 10.1016/j.ijheatmasstransfer.2015.01.048.

[5] H. Saleh, A. I. Alsabery, M. A. Sheremet, M. Ghalambaz, and I. Hashim. Impact of fluid–structure interaction on unsteady flow within wavy-heated wall cavity. *International Journal of Numerical Methods for Heat & Fluid Flow*, 35(4):1484–1511, 2025. doi: 10.1108/HFF-01-2025-0045.

[6] Vipin Gupta and M. S. Barak. Photo-thermo-piezo-elastic waves in semiconductor medium subject to distinct two temperature models with higher order memory dependencies. *International Journal of Numerical Methods for Heat & Fluid Flow*, 34(1):84–108, 2024. doi: 10.1108/HFF-07-2023-0380.

[7] Vivek Kumar, Ashwani Assam, and Deepu Prabhakaran. Nonlinear thermo-fluid–structure interactions of two interacting flexible cantilever plates in a differentially heated cavity. *Physics of Fluids*, 38(9):092128, 2026. doi: 10.1063/5.0349114.

[8] Bruno A. Boley and Jerome H. Weiner. *Theory of Thermal Stresses*. John Wiley and Sons, New York, 1960. Duhamel–Neumann law, thermoelastic pressure coefficient, free thermal curvature, coupled energy term.

[9] Stefan Turek and Jaroslav Hron. Proposal for numerical benchmarking of fluid-structure interaction between an elastic object and laminar incompressible flow. In *Fluid–Structure Interaction*, volume 53 of *Lecture Notes in Computational Science and Engineering*, pages 371–385. Springer, 2006. doi: 10.1007/3-540-34596-5_15. Transient large-displacement benchmark; not applicable to the present steady small-strain formulation (cited to state that restriction).

[10] Franco Brezzi and Michel Fortin. *Mixed and Hybrid Finite Element Methods*, volume 15 of *Springer Series in Computational Mathematics*. Springer-Verlag, New York, 1991. doi: 10.1007/978-1-4612-3172-1. Inf–sup stability of mixed velocity–pressure discretizations.

[11] H. S. Carslaw and J. C. Jaeger. *Conduction of Heat in Solids*. Clarendon Press, Oxford, 1947. Steady slab conduction and conjugate flux-continuity formulation.

[12] Marko Draskic, Jerry Westerweel, and Rene Pecnik. The stability of stratified horizontal flows of carbon dioxide at supercritical pressures. *Journal of Fluid Mechanics*, 1012:A17, 2025. doi: 10.1017/jfm.2025.10162.

[13] M. Lesoinne and C. Farhat. Geometric conservation laws for flow problems with moving boundaries and deformable meshes, and their impact on aeroelastic computations. *Computer Methods in Applied Mechanics and Engineering*, 134:71–90, 1996. doi: 10.1016/0045-7825(96)01028-6. Discrete GCL requirements; motivates the steady deformed-domain restriction of the present baseline.

[14] A. I. Alsabery, H. Saleh, M. Ghalambaz, A. J. Chamkha, and I. Hashim. Fluid-structure interaction analysis of transient convection heat transfer in a cavity containing inner solid cylinder and flexible right wall. *International Journal of Numerical Methods for Heat & Fluid Flow*, 29(10):3756–3780, 2019. doi: 10.1108/HFF-10-2018-0593.

[15] Ivan C. Christov, Vincent Cognet, Tanmay C. Shidhore, and Howard A. Stone. Flow rate–pressure drop relation for deformable shallow microfluidic channels. *Journal of Fluid Mechanics*, 841:267–286, 2018. doi: 10.1017/jfm.2018.30.

[16] Stephen P. Timoshenko and James M. Gere. *Theory of Elastic Stability*. McGraw-Hill, New York, 2 edition, 1961. Pinned–pinned column/plate-strip Euler load and shear-deformation correction estimates.

[17] Stephen P. Timoshenko and James N. Goodier. *Theory of Elasticity*. McGraw-Hill, New York, 3 edition, 1970. Plane-strain closure, equilibrium formulation, plate-strip flexural rigidity.

[18] Peyman Rahimimotlagh, Mohammad Ahmadian-Elmi, and Mohammad Reza Hajmohammadi. Numerical investigation of conjugate heat transfer in a flattened cross-sectional thick channel. *International Journal of Thermal Sciences*, 195:108637, 2024. doi: 10.1016/j.ijthermalsci.2023.108637.